

Research Report

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Building a Defense-in-Depth Biosecurity Strategy for the AI Era

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About This Report

The ongoing development of artificial intelligence (AI) tools for biotechnology has the potential to yield many benefits, but it might also introduce significant risks arising from misuse. To address this, we conducted a comprehensive analysis of diverse pathways to harm that are available to priority threat actors. Our analysis led us to identify a set of nine priority mitigations that can meaningfully safeguard the world from the most severe AI-enabled biological weapons. In this report, we introduce a comprehensive mitigation strategy consisting of mutually reinforcing interventions that show significant promise for preventing a high-consequence AI-enabled biological attack by collectively managing access to dangerous and dual-use information, disrupting materials access, deterring nefarious activity, and detecting misuse signals proactively. The mitigation strategy demonstrates how prevention must evolve beyond access controls alone: Effectiveness depends on complementary detection architectures and deterrence mechanisms to meaningfully reduce risk across all possible misuse pathways. We analyze the mitigation strategy's strengths and weaknesses, describe areas of mutual reinforcement, and propose next steps for how policymakers and decisionmakers in the public and private sectors can work collaboratively to advance this strategy.

Center on AI, Security, and Technology

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Summary

The convergence of artificial intelligence (AI) and biology offers great promise for such fields as medicine and public health, and it promises the potential to yield transformative societal benefits. Yet this same convergence could introduce serious risks to global health and national security. As capabilities continue to advance, AI-enabled biotechnology might lower the technical, operational, and motivational barriers to creating biological weapons. It might also enable devastating attacks involving enhanced or novel pathogens, including events far worse than historical pandemics that caused many millions of fatalities globally. The dual-use nature of biotechnology requires proactive mitigation measures to counter these high-consequence threats.

Therefore, we undertook an effort to identify which mitigation measures that have been proposed to date provide the greatest protection in aggregate against the most serious AI-enabled biological incidents, which are those enabled by human-transmissible pathogens. There has been limited systematic analysis of how mitigations interact across the biological risk chain, how they address different actor profiles, and how robust they remain under divergent AI futures. With this report, we seek to move beyond cataloging options and toward structuring them into an integrated prevention framework.

To develop a mitigation strategy that is robust, complementary, and deployable across a wide variety of threat scenarios, we considered each essential step of the bioweapon-development pathway prior to an attack, from ideation to weaponization, and assessed the capabilities of various actor types to achieve such weapons. We then examined how different prototype networks of mitigations added friction to five primary milestones that nefarious actors must bypass to successfully execute an attack using AI-enabled biological weapons. These milestones are (1) developing and sustaining motivation to cause harm using biological means, (2) using the appropriate tools and data to ideate and design an enhanced or novel biological threat, (3) accessing the right information and expertise to successfully execute the steps to biological threat development and weaponization, (4) obtaining the appropriate precursor materials and equipment to translate designs to physical manifestations, and (5) circumventing biosecurity controls and law enforcement to avoid detection.

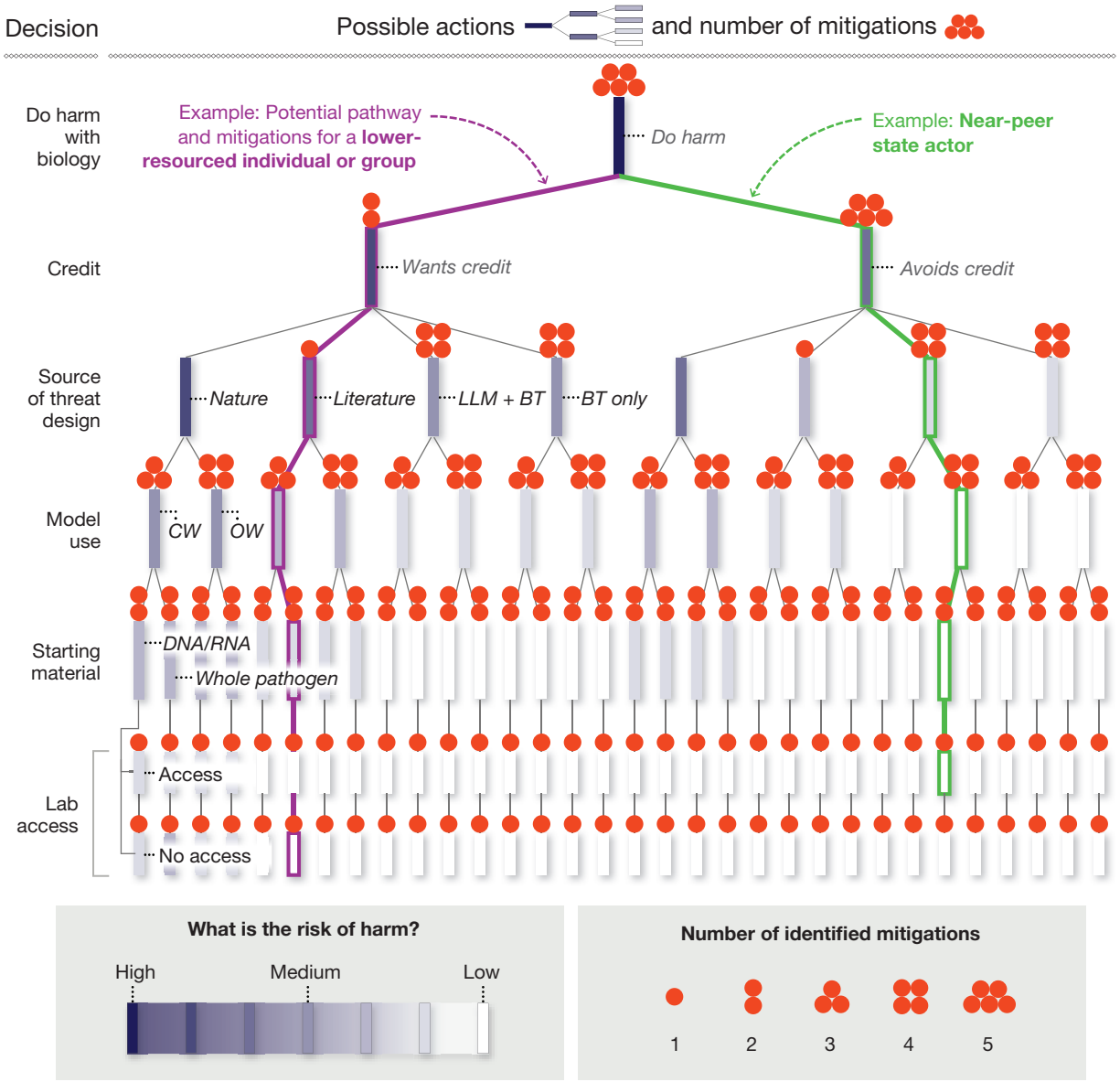
Next, we further evaluated these prototype networks of mitigations against representative threat actor profiles to stress test the network and determine the interplay between different mitigations. A central finding from this analysis is that different actor profiles are susceptible to different classes of mitigations, such that coverage across the different harm pathways is necessary to create defense-in-depth that meaningfully prevents misuse. Some actors appear relatively amenable to disruption but are difficult to deter, while others appear difficult to disrupt but are more susceptible to deterrence. This trade-off motivates the creation of a layered strategy that combines approaches rather than one that relies on any single intervention at any single node to achieve defense-in-depth that prevents harm across several pathways (see Figure S.1).

Our findings suggest that a strategy that pairs disruption-oriented mitigations with credible deterrence- and detection-oriented backstops is more robust than one that concentrates investment at any single point. In-depth discussion of the analyses and stress testing that led to the selection of these nine mitigations can be found in Chapter 3.

From our assessment, nine priority mitigations emerged to form a layered, defense-in-depth protection approach, as outlined in Table S.1. This strategy has dense coverage across early stages of the biological weapons risk pathway and a central focus on the design-build-test-learn cycle primarily because we seek to reduce actors' capacities to effectively design a novel pathogen that could initiate global-scale consequences. However, none of these early-stage mitigations will be perfectly implemented: Information controls can be circumvented or are irrelevant for expert actors, synthesis screening has coverage gaps, and access-controlled platforms can be exploited by credentialed insiders or sufficiently resourced actors. In these cases, disruption shifts to additional layers, such as the proactive detection of misuse behavior across the risk chain and the

FIGURE S.1

Example of Layered Defense Achieved by This Report's Mitigations



NOTE: BT = biological tool; CW = closed weight; DNA = deoxyribonucleic acid; LLM = large language model; OW = open weight; RNA = ribonucleic acid.

deterrence of nefarious activity. Taken together, these mitigations are designed to maximize the upsides of tomorrow's innovations without enabling pandemic-scale harm.

We detail these mitigations in this report along with key considerations for their adoption and implementation. In Chapter 4, we provide an outline of how policymakers and decisionmakers in the public and private sectors can work collaboratively to advance this mitigation strategy for preventing global-scale AI-enabled biological incidents.

TABLE S.1
Summary of the Mitigation Strategy

Mitigation Class	Mitigation
Manage access to dangerous and dual-use information	• Development of tamper-resistant safeguards for open-weight AI models • Protection of sensitive biological data • Access-control platforms and mechanisms for dual-use AI systems
Disrupt materials access	• Biological precursor screening
Deter nefarious activity	• Pathogen early-warning systems • Bioattribution and forensics • Demonstrated rapid-response and consequence-mitigation capabilities
Detect misuse signals proactively	• Real-time monitoring and reporting of misuse of closed-weight general-purpose AI models • Centralized AI x Bio information-sharing nexus

NOTE: AI x Bio = the intersection of AI and biology.

Key Findings

- **Different threat actors require different prevention mechanisms.** Resource-constrained individuals might be thwarted at material or information access chokepoints. However, technologically sophisticated or well-resourced groups, such as state actors, might be immune to access controls and therefore require deterrence through elevated perceived costs and a degraded expected utility of attack.
- **Prevention must evolve beyond access controls alone.** Although restricting dual-use information and materials remains necessary, determined actors with sufficient resources can circumvent these barriers. An effective strategy, therefore, requires a complementary detection architecture that identifies and disrupts misuse patterns in digital and physical domains before they culminate in harm, shifting the paradigm from denial to integrated monitoring and early response.
- **Aggregate signals matter.** Threat actors operating across multiple nodes of the AI-biology ecosystem—probing different models, querying multiple synthesis providers, sourcing materials from separate vendors—produce signals that appear ambiguous when observed individually but form identifiable patterns of concern when analyzed collectively. Realizing this aggregate value requires centralized AI x Bio information-sharing infrastructure that no single entity can provide alone.
- **Invest early.** Implementing this strategy's mitigations will require investments in new research, institutional adaptation, legal clarification, and international coordination. Waiting until more-advanced biological threats are undeniable will be waiting too long.

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CHAPTER 1

Introduction and Background

Purpose of This Report

This report presents a comprehensive prevention and mitigation strategy aimed at reducing the likelihood of artificial intelligence (AI)-enabled biological incidents by reinforcing barriers before a harmful biological agent can be designed, developed, tested, and/or released. Our analysis focuses on the intersection of advanced general-purpose AI (GPAI) systems, AI-enabled biological tools (BTs), and biological research, development, and operational workflows. With this prevention-oriented mitigation strategy, we propose a layered and integrated combination of interventions intended to extend time, increase effort, and raise the probability of detection, disruption, or deterrence prior to release of a biological weapon.

The scope of this report is deliberately narrow.[1] It focuses on *high-consequence AI-enabled biological incidents*, which we define as biological events capable of producing impacts on the order of ten times the scale of the coronavirus disease 2019 (COVID-19) pandemic or greater, societal disruption, or systemic destabilization. Within that framing, we have prioritized the detection, disruption, or deterrence of the deliberate weaponization of biological agents with enhanced or novel characteristics (e.g., transmissibility, pathogenicity, latency, immune evasion) that can infect humans.[2] We map the characteristics of prioritized threat actors and consider interventions that will prevent AI-enabled biological incidents from occurring rather than post-release responses, which might suffer from offense-defense asymmetries in effectiveness.[3] Such mitigations aim to strengthen existing technical, operational, and motivational barriers to prevent biological attacks from occurring.

We neither attempt to provide a comprehensive biosecurity framework across all biological risks nor offer an exhaustive inventory of technical safeguards. Instead, we map the AI-enabled biological risk chain and then assess how varying actor characteristics and profiles influence potential misuse pathways, and we use this analytical foundation to develop a set of integrated mitigation measures. This work is urgent for several reasons. First, AI capabilities are advancing rapidly, and improvements in reasoning, multimodal integration, tool use, and agentic capabilities will have downstream implications for biological design, scientific workflows, and laboratory experimentation. Although the precise trajectory of AI development is uncertain,

[1] Mitigations for AI-enabled biological attacks of other sizes and severities are addressed by complementary lines of effort at RAND that overlapped with the timeline of this research. See Aurelia Attal-Juncqua, John P. Tarangelo, Evan Davison Kotler, Casey Aveggio, Marie Burns, Barbara Del Castello, Tyler Burns, Steph Guerra, and Saskia Popescu, *Building Resilience Against Artificial Intelligence–Enabled Biological Threats: Conference Proceedings from a RAND and Helena Workshop*, RAND Corporation, CF-A4954-1, 2026.

[2] A potential exception is the deliberate weaponization of naturally occurring pathogens, such as smallpox, which could plausibly reach global-scale impact thresholds under certain conditions. However, given the eradication of smallpox worldwide, the centralized and close monitoring of laboratory stocks, and the challenges of synthesizing this virus de novo, we believe that the mitigations in this strategy will also provide protection against smallpox weaponization.

[3] See Casey Aveggio, Aman J. Patel, Sella Nevo, and Kyle Webster, *Exploring the Offense-Defense Balance of Biology: Identifying and Describing High-Level Asymmetries*, RAND Corporation, PE-A4102-1, August 2025.

capabilities are evolving in ways that could affect both legitimate research and potential misuse, such as by improving protocol generation, experimental planning, troubleshooting, and integration with laboratory or procurement tools. Second, although numerous mitigation strategies have been enumerated in prior literature and policy discussions, these efforts have often been presented as lists of discrete controls rather than as components of a coherent and reinforcing strategy. And third, AI-native technical safeguards and real-world mitigations take time to develop, test, and implement. Technical safeguards for AI systems, screening mechanisms for biological materials, access controls for sensitive biological data, and cross-sector information-exchange frameworks require research, institutional adaptation, legal clarity, and, in many cases, international coordination. Delaying prioritization until AI capabilities mature to undeniable uplift and novelty further risks widening the gap between technological advancement and protective infrastructure.

This report is intended for policymakers who are responsible for national security, public health, and science and technology governance; leaders within AI companies and biological research institutions; and cross-sector stakeholders involved in compliance, oversight, and international coordination. For policymakers, the report provides a structured framework for understanding how (1) advanced AI systems intersect with biological risk pathways and (2) a system of preventive interventions might be most impactful in terms of preventing AI-enabled biological incidents that are based on priority threat actor capabilities and intentions. For AI developers and laboratory leaders, this report offers guidance on how safeguards can be embedded into technical and operational workflows in ways that align with prevention objectives for and areas of complementary investment in off-model mitigations. For interagency and international audiences, the report outlines areas in which coordination, information exchange, and shared standards are likely to reduce the possibility of misuse.

Public- and private-sector buy-in are needed to implement the mitigation strategies in this report. However, sustained commitment from the U.S. government is a critical enabling factor to avoid an overreliance on voluntary industry actions. Overreliance on voluntary industry action might lead to a "race to the bottom" in which even a floor of requirements is improperly executed. Federal leadership, investment, and policymaking are necessary for the mitigation strategies to reach effectiveness at scale.

Key Context

AI and Emerging Biological Risks

In this report, *AI* refers broadly to machine learning–based systems, including both established approaches used in biological research and more-recent advances in foundation models and AI agents. These systems play different roles across the biological workflow. General-purpose models support reasoning and design tasks, agents can coordinate multistep processes, and domain-specific BTs enable such tasks as protein-structure function prediction or sequence analysis. Increasingly, these components are being combined within integrated workflows that accelerate biological research and development.

Advances in these systems and tools over the past several years have expanded the variety of tasks that can be automated, accelerated, or optimized across scientific and operational domains.[4] In biology, such advanced AI systems can assist with protein-structure prediction,[5] genome modeling, experimental plan-

[4] Cyndy S. Groff-Vindman, Benjamin D. Trump, Christopher L. Cummings, Madison Smith, Alexander J. Titus, Ken Oye, Valentina Prado, Eyup Turmus, and Igor Linkov, "The Convergence of AI and Synthetic Biology: The Looming Deluge," *npj Biomedical Innovations*, Vol. 2, July 1, 2025.

[5] Lingtao Chen, Qiaomu Li, Kazi Fahim Ahmad Nasif, Ying Xie, Bobin Deng, Shuteng Niu, Seyedamin Pouriyeh, Zhiyu Dai, Jiawei Chen, and Chloe Y. Xie, "AI-Driven Deep Learning Techniques in Protein Structure Prediction," *International Journal of Molecular Sciences*, Vol. 25, No. 15, August 1, 2024.

ning, workflow optimization, and large-scale information synthesis.[6] These developments offer substantial benefits for medicine, biotechnology, and research productivity.

At the same time, the integration of AI into biological research environments surfaces new potential risks. Indeed, AI systems can reduce knowledge requirements, accelerate experimental iteration cycles, and shift aspects of biological insight into earlier, digital stages of the development process, during which oversight mechanisms might be weaker.[7] AI systems can also support coordination, planning, and information acquisition in ways that expand the number of actors who could feasibly achieve a successful biological attack. For GPAI systems, this often represents a redistribution of capability rather than a wholly new category of risk across existing threat pathways. However, biological design tools might introduce meaningful novelty into the risk landscape—introducing unprecedented capabilities that existing frameworks were not designed to address. These dynamics might become more pronounced as laboratory automation, cloud-based experimentation platforms, and AI-integrated research environments continue to develop. For example, cloud laboratories and other remotely or autonomously operated experimental platforms could allow users to move from digital design to physical testing without being physically embedded in a laboratory, while automation of routine experimental steps could reduce a reliance on tacit bench expertise for setup, execution, and iteration.[8] Integrated systems that combine model-assisted design, protocol generation, materials ordering, and experiment management could also redistribute parts of the workflow that once depended on highly trained researchers across a wider variety of users. In such settings, the boundary between digital design and physical execution might become increasingly compressed, reducing reliance on tacit expertise and shifting key control points earlier in the workflow, including toward platform access, upstream design choices, and procurement pathways. This evolution could both amplify efficiency gains and introduce new challenges for oversight and intervention, particularly when end-to-end processes become more accessible within integrated or remotely accessible systems.

Approach

Theory of Change

The underlying theory of change for this strategy relies on the assumption that any actor seeking to execute a pandemic-scale biological attack must undertake a series of steps across the biological risk chain, and they often must do so with technical, motivational, and operational barriers that must be overcome. These barriers include acquiring relevant information, accessing or generating biological materials, designing or enhancing a threat, translating digital outputs into physical systems, and avoiding detection or attribution.[9]

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[6] Kang Zhang, Xin Yang, Yifei Wang, Yunfang Yu, Niu Huang, Gen Li, Xiaokun Li, Joseph C. Wu, and Shengyong Yang, "Artificial Intelligence in Drug Development," *Nature Medicine*, Vol. 31, January 2025.

[7] Jaspreet Pannu, Doni Bloomfield, Robert MacKnight, Mortiz S. Hanke, Alex Zhu, Gabe Gomes, Anita Cicero, and Thomas V. Inglesby, "Dual-Use Capabilities of Concern of Biological AI Models," *PLoS Computational Biology*, Vol. 21, No. 5, May 8, 2025; National Academies of Sciences, Engineering, and Medicine, "AI-Enabled Biological Design and the Risks of Synthetic Biology," in *The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations*, National Academies Press, 2025a.

[8] Ying-Chiang Jeffrey Lee, Bria Persaud, Barbara Del Castello, Allison Berke, and Gustavs Zilgalvis, *Documenting Cloud Labs and Examining How Remotely Operated Automated Laboratories Could Enable Bad Actors*, RAND Corporation, PE-A3851-1, April 2025; Nazish Jeffery, Sarah R. Carter, Tessa Alexanian, Oliver Crook, Samuel Curtis, Richard Moulange, Shrestha Rath, Sophie Rose, and Jennifer Clarke, "Bio x AI: Policy Recommendations for a New Frontier," Federation of American Scientists, December 12, 2023.

[9] Michael A. Washington, Joubert Paulino, and Siang-Chean Kua, "Barriers to Biological Weapons Development: Potential Implications for Pathway Disruption," *Countering WMD Journal*, No. 28, Spring/Summer 2024; David Luckey,

By layering mitigations at multiple points along this chain, it is possible to extend the time required for malicious activity, deter potential nefarious activity or intentions, increase the operational effort necessary to proceed, and raise the likelihood that suspicious behavior is detected and disrupted before release. Actors vary in expertise, resources, motivation, and risk tolerance, and as AI systems evolve, these differences interact with threat pathways in ways that can lower barriers to misuse. Therefore, this strategy considers threat pathway characteristics, actor profiles, and areas of AI uplift when identifying points of intervention.

Scope

This strategy is explicitly scoped to high-consequence AI-enabled biological incidents with projected impacts of at least ten or more times the order of magnitude of the COVID-19 pandemic, including scenarios involving tens to hundreds of millions of fatalities globally alongside severe economic disruption, health-system collapses, and prolonged societal impacts. Within this framing, we conducted an analysis that prioritized threats that infect humans, are transmissible, and are enhanced or novel in ways that reduce the likelihood of natural occurrence. Although our analysis focused on pathogens that infect humans, many of the mitigations that we outline in the strategy are also applicable to preventing biological misuse targeting plants and animals. The focus on worst-case scenarios reflects the view that planning for extreme outcomes strengthens safeguards that are also relevant for most lower-severity risks.

The mitigations considered in this report span multiple intervention types and points in the risk chain, including technical and model-layer safeguards, access and deployment controls, upstream governance and monitoring mechanisms, and selected digital-to-physical chokepoints. The emphasis of this report is on prevention of AI-enabled biological incidents before a pathogen's release. Mitigations are therefore concentrated "left of boom" with the objective of strengthening technical, operational, and motivational barriers that reduce the probability that an attack can be successfully executed. Although some of the mitigations included in this report have post-release benefits, we selected them primarily because of their deterrent qualities rather than for their downstream effects. Post-release response measures are critical components of broader bioresilience and emergency responses, but they are not the central focus of this strategy, which aims to prevent a pathogen's release in the first place given the challenge of containing and responding to a pandemic-level event using biological offense and defense.[10]

AI-Enabled Biothreat Risk Pathway

To support the development of a layered prevention strategy, we used a simplified risk pathway to describe the key steps that a threat actor must complete to develop a biological weapon. This framing allowed us to assess where we can apply mitigations across the full risk chain and how they can reinforce one another. Although we depict the pathways linearly for clarity, real-world pathways are often iterative rather than strictly sequential. The risk chain that we use in this report integrates insights from terrorism planning, biological weapons

Sara Duhachek Muggy, Taylor Frey, David Stebbins, Tracey Rissman, Bianca Espinosa, Daniel Tapia, Greg McKelvey, Jr., Neeti Pokhriyal, Joseph Dawson, Sara Hughes, Morgan Sandler, Rushil Bakhshi, Marta Kepe, Geoffrey Kirkwood, Sarah W. Denton, David DeSmet, Minami Makino, Ella Guest, Sina Beaghley, Suzanne Genc, Michael Miller, Skye A. Miner, Barbara Del Castello, Forrest W. Crawford, Jeffrey Lee, Clay Strickland, Sunny D. Bhatt, John Vahedi, Lydia Grek, Vanya Barrer, Ramiro Insuasti, Jr., Jack Lashendock, Derek Roberts, Aleksandr Esparza Hartunian, Shannon Walsh, Will Shumate, Elliott Brennan, Tyler Liggett, Kara Jia, Ajay K. Kochhar, James Smith, and James Ryseff, *Mitigating Risks at the Intersection of Artificial Intelligence and Chemical and Biological Weapons*, Homeland Security Operational Analysis Center operated by the RAND Corporation, RR-A2990-1, 2025; Tracey Rissman and Annette Prieto, *Attributing Biological Weapons Use: Strengthening Department of Defense Capabilities to Investigate Deliberate Biological Incidents*, RAND Corporation, RR-A2360-1, 2024.

[10] Aveggio et al., 2025.

development, and synthetic biology, and it highlights how the use of AI might increase the likelihood of success by accelerating tasks or lowering technical barriers.[11]

The framework focuses on seven stages: intention, ideation, design, build, test, learn, and weaponization. Together, these stages capture the progression from initial motivation to a deployable capability, which is depicted in Figure 1.1. We exclude the "release" stage in this risk pathway given this strategy's focus on left-of-boom mitigations to prevent a release. Each stage represents a potential point for intervention. This strategy, therefore, assesses how AI-enabled tools create uplift across these stages and identifies where on the pathway that mitigations can most effectively reduce risk. Further detail on the stages of the AI-enabled biological risk pathway,[12] including stage definitions and AI-specific impacts, is provided in Appendix B.[13]

Actor Characteristics

Harmful outcomes typically require the convergence of several factors: intent, access to relevant tools or infrastructure, the capability to use those tools effectively, and the opportunity to act without detection or disruption. Intent or motivation alone is often a poor predictor of risk because actors might be highly motivated yet lack the expertise, resources, or operational capacity to act, while those who are moderately motivated but have greater capability and access might pose a more-immediate threat. In the context of AI-enabled biology, risk therefore emerges when these factors align, particularly as AI systems might redistribute capabilities across actors in ways that lower barriers to effective misuse. Furthermore, actors differ significantly in their abilities to exploit AI-enabled biological capabilities, reflecting variation in how intent, capability, access, and opportunity combine in practice. Table 1.1 summarizes the primary characteristics that influence an actor's ability to leverage AI-enabled biological capabilities.[14]

Using the capability dimensions described above, we mapped threat actor profiles by analyzing how different types of actors fulfill these key capability dimensions that are relevant to misuse success. Figure 1.2 summarizes how these characteristics vary across several actor profiles that are relevant to AI-enabled biological risk, and it provides an assessment of each of these dimensions at a baseline without the influence of AI uplift. The levels are qualitative assessments derived through iterative team discussions and are intended to illustrate relative differences in capability and constraints across actors rather than precise absolute measurements. It is important to note that these capability dimensions are not static but can be shifted by interactions with and the capabilities of advanced AI systems. We used Figure 1.2 to evaluate the robustness of our proposed mitigation strategies; it informed how effective specific controls would be against different actors.

[11] Roger Brent and Greg McKelvey, Jr., *Contemporary Foundation AI Models Increase Biological Weapons Risk*, RAND Corporation, PE-A3853-1, December 2025; Daniel M. Gerstein, *Bioterror in the 21st Century: Emerging Threats in a New Global Environment*, Naval Institute Press, 2009, p. 156; Office of the Director of National Intelligence, "JCAT Counter Terrorism Guide for Public Safety Personnel," webpage, undated; Sophie Rose and Cassidy Nelson, *Understanding AI-Facilitated Biological Weapon Development*, Centre for Long-Term Resilience, October 2023; U.S. Army, "Army Antiterrorism Officer Ready Reference: Vigilance," undated.

[12] Rose and Nelson, 2023.

[13] National Academies of Sciences, Engineering, and Medicine, "Design-Build-Test-Learn: Impact on the Synthetic Biology Process," in *The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations*, National Academies Press, 2025b; Kyle Brady, Jeffrey Lee, Dawid Maciorowski, Alyssa Worland, Jordan Despanie, Bria Persaud, Barbara Del Castello, Henry Alexander Bradley, Grant Ellison, Charles Teague, Sarah L. Gebauer, Greg McKelvey, Jr., Steph Guerra, and Ella Guest, *Bridging the Digital to Physical Divide: Evaluating LLM Agents on Benchtop DNA Acquisition*, RAND Corporation, RR-A4591-1, 2026.

[14] Navigating the Benefits and Risks of Publishing Studies of In Silico Modeling and Computational Approaches of Biological Agents and Organisms—A Workshop, National Academies of Sciences, Engineering, and Medicine, April 3–4, 2025.

FIGURE 1.1

Stages of the AI-Enabled Biological Threat Risk Pathway

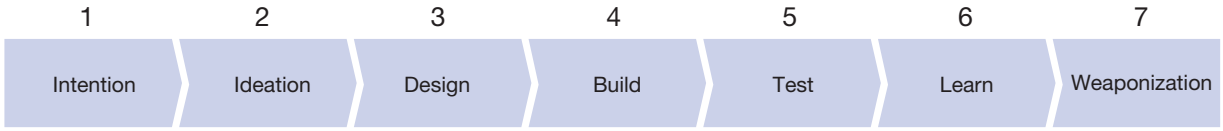
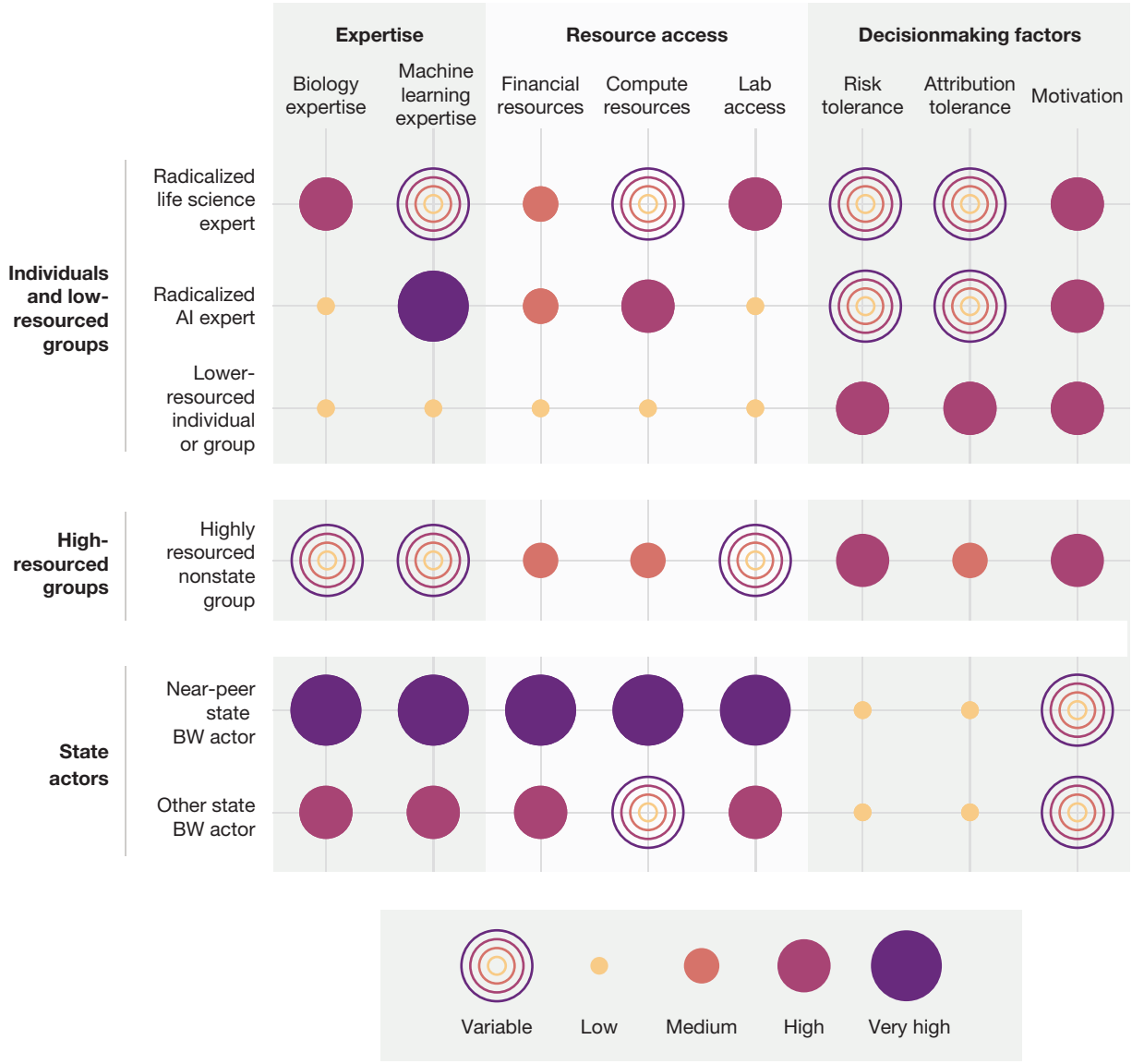


TABLE 1.1

Key Factors Shaping Actor Capability in AI-Enabled Biological Misuse

Actor Capability Dimension	Description	Relevance for AI-Enabled Risk
Biological expertise	Training and experience in biological research, pathogen biology, or laboratory experimentation at an undergraduate level or higher	Influences the ability to develop a biological agent with or without meaningful AI assistance and to interpret AI outputs and translate computational designs into viable biological systems
Computational and AI expertise	Familiarity with machine learning tools, model manipulation, or computational biology workflows	Enables actors to more effectively leverage AI tools for biological design, integrate model outputs into biological workflows, overcome model safeguards, and/or train or fine-tune models that are customized for nefarious purposes
Financial resources	Funding to support experimentation or compute-intensive workflows	Expands the scale and persistence with which actors can pursue biological activities, particularly without engaging with providers or resources that require legitimate access
Access to compute	Availability of compute on premises or through a cloud provider determines access to deep inference and potential fine-tuning using advanced AI models or specialized biological design tools	Shapes the extent to which actors can exploit AI-enabled biological design or analysis to develop an enhanced or novel biological weapon, particularly through compute-intensive training, fine-tuning, or inference
Access to laboratory infrastructure	Direct or indirect availability of laboratory space, experimental equipment, synthesis services, or biological materials	Determines whether actors can move from digital design to physical biological experimentation with limited operational friction
Risk tolerance	Willingness to accept personal or organizational harm in the process of designing or developing an enhanced or novel biological threat	Influences the likelihood that actors attempt higher-risk biological activities outside appropriate safety controls
Attribution tolerance	Willingness to accept personal, legal, or operational risks associated with illicit activity (e.g., willingness to be caught and held accountable)	Influences the likelihood that actors attempt to perform a biological attack, should high confidence detection and attribution capabilities exist
Motivation and time horizon	Ideological, political, or strategic motivations and the time frame over which actors pursue objectives	Shapes the persistence and willingness to invest effort in overcoming technical barriers

FIGURE 1.2
Relative Capability Dimensions of Actors at Baseline Prior to Potential AI Uplift



NOTE: BW = biological weapon.

Priority Actor Profiles

Building on the capability dimensions described previously, we identified several actor profiles that might be particularly relevant in the context of AI-enabled biological risks.[15] These profiles represent illustrative

[15] Barbara Del Castello and Henry H. Willis, *Assessing the Impacts of Technology Maturity and Diffusion on Malicious Biological Agent Development Capabilities: Demonstrating a Transparent, Repeatable Assessment Method*, RAND Corporation, RR-A3662-1, 2025; Adeline E. Williams, Barbara Del Castello, Jeffrey Lee, Derek Roberts, John P. Tarangelo, Jay Atanda, Alejandro Colman-Lerner, Jeff Gerold, and Roger Brent, *Developing a Risk-Scoring Tool for Artificial Intelligence–Enabled Biological Design : A Method to Assess the Risks of Using Artificial Intelligence to Modify Select Viral Capabilities*, RAND Corporation, RR-A4490-1, 2026.

combinations of expertise, access to resources and infrastructure, and motivational factors that could enable actors to exploit AI-enabled biological capabilities. The profile categories are as follows:

- a radicalized life science expert who possesses biological expertise and access to laboratory space and materials
- a radicalized AI expert with access to advanced AI capabilities and computational resources
- a lower-resourced individual or group—a nonstate lone wolf or group that is highly motivated but lacks the AI and life science expertise of the other nonstate actors
- a highly resourced nonstate group—a motivated nonstate group with members that have expertise in AI and biology
- a near-peer state actor—a state actor with AI and life science research capabilities on par with, or close to, those of the United States
- another state actor—a state actor with AI and/or life science research capabilities that are substantially less advanced than those of the United States.

These profiles represent distinct and priority combinations of expertise, access, and motivation that might shape how actors interact with AI-enabled biological capabilities. We recognize that actors within these profile categories might have nuanced differences. However, to keep our analysis feasible, we ascribed one level of expertise, access, and motivation for each actor profile, as well as a single description of actor motivations for each profile category. This analysis of likely actors revealed an encouraging trade-off in how we developed our final strategy: It is very challenging to detect or disrupt some actors (e.g., highly capable state actors), but they seem very deterrable, while there is a potential to use access controls and monitoring to detect or disrupt other actors, but these actors are more challenging to deter (e.g., they are ideologically motivated nonstate actors with extremist tendencies). This realization motivates a multilayered approach because different actors are stopped by different mitigations, but it also means that, although no mitigation is a silver bullet capable of thwarting all actors, one could develop an effective mitigation strategy that addresses all likely actor profiles.

Mitigation Strategy Development and Evaluation

Our goal was to identify a focused set of mitigations that could (1) meaningfully reduce the risk of AI-enabled biological incidents by diverse threat actors and (2) be implemented within the next two to five years by governments and industry. Rather than proposing a broad set of actions, we prioritized developing a concise and complementary network of mitigations. We began by compiling an initial set of 67 candidate mitigations from academic and gray literature, including prior RAND analyses. These mitigations spanned multiple layers of defense, including technical safeguards on AI systems, controls on material and data access, user access restrictions, and policy-oriented interventions. We then conducted an iterative refinement process to evaluate how these mitigations interact, their relative strengths, and their performance across a variety of plausible AI-enabled biological scenarios. This included assessing how different combinations of mitigations could reduce risk, identifying complementarities and redundancies, and examining the mitigations' effectiveness across varying actor types, threat types, and pathways for material and information acquisition. We also mapped mitigations across the AI-enabled biothreat risk chain to identify key points of intervention.

This process resulted in a refined set of nine core mitigations. We provide additional detail on the methodology—including scenario dimensions, evaluation criteria, and the mitigation refinement process—in Chapter 3 and Appendix C.

Limitations

Several limitations shaped our analysis. Uncertainty in the trajectory of AI development makes precise forecasting difficult, and future advances might shift the relative importance of specific mitigations. The safeguards prioritized in this report reflect existing capabilities and near-term expectations, and they might require revision as technologies evolve.

The assumption guiding this strategy is that, although some actors might be more difficult to deter and others more difficult to disrupt, every actor faces some level of constraint that can be strengthened through appropriately layered mitigations. The objective is not to eliminate all risk but to alter threat actors' cost-benefit calculations and increase the probability of early detection, deterrence, and/or disruption.

Finally, we provide limited treatment of post-release harm mitigation given the report's prevention-focused scope. This is an important focus because of several asymmetries that are likely to favor attackers over defenders (kinetic consideration, financial burden, threat surface, and consequences of failure).[16] This boundary is deliberate and reflects a decision to concentrate analytical attention on reducing the probability of pandemic-scale, AI-enabled incidents occurring in the first place while recognizing that broader response and resilience measures remain essential complements to biodefense efforts.

[16] Aveggio et al., 2025.

CHAPTER 2

Strategy Overview

Using our analysis of the various requirements to achieve harm along the biological misuse risk pathway (acquiring dangerous information, accessing biological materials, designing or enhancing a pathogen, and avoiding detection or attribution),[1] we propose a network of nine overlapping and reinforcing interventions as part of our prevention-focused mitigation strategy. These mitigations aim to strengthen technical, operational, and motivational barriers to ensure that a diverse set of threat actors will be unsuccessful at planning and executing an AI-enabled biological attack at a pandemic-level scale. We organized the interventions into four mitigation classes to ensure the strategy's robustness: manage access to dangerous and dual-use information, disrupt materials access, deter nefarious activity, and detect misuse signals proactively. Each mitigation was assessed as having varying levels of effectiveness across different priority threat actors and risk pathways. Many mitigations are also mutually reinforcing, meaning that joint implementation enhances their effectiveness. There is no silver bullet mitigation that is expected to be a bottleneck across every possible pathway to misuse. Therefore, strengthening a network of mitigations across the risk chain and routinely evaluating and updating the strategy are necessary to meaningfully reduce the risk of misuse. A summary table of the mitigations is in Appendix A.

Mitigation Strategy Assumptions

Each mitigation section in this chapter includes a set of explicit assumptions that define the conditions under which that mitigation is expected to be effective. These assumptions provide a transparent causal logic of the mitigation itself by identifying which threat actor profiles it is designed to address, what technological and regulatory conditions it depends on, and where its effectiveness is likely to erode if circumstances change. For example, should open-weight models fall permanently behind in biological capability compared with frontier closed-weight models, the first mitigation in this chapter might not be worth prioritization. Stating these assumptions explicitly makes each mitigation empirically testable. Each set of assumptions should be treated as live hypotheses that stakeholders should actively probe and track. Evidence that a core assumption is failing—whether because it was wrong at the outset or because of shifting technologies, threat landscapes, or governance ecosystems—should trigger reassessment of that mitigation's design, scope, or prioritization. The assumptions are meant to serve as an analytical tool and an accountability mechanism to maximize the strategy's effectiveness over time.

[1] For more details on this analysis, see Chapter 3 and Appendixes B and C.

Manage Access to Dangerous and Dual-Use Information

Actors of various sophistication and resource levels might find opportunities to use GPAI models and AI-enabled BTs to uplift their ability to successfully execute tasks across the entire biological risk chain, from intention to weaponization. This uplift could take several forms. For example, it could manifest as technical information that is not otherwise accessed efficiently or comprehensively, as laboratory troubleshooting or coaching that is typically reserved for an expert scientist, or as operational planning related to biosecurity circumvention or a selection of the most impactful pathogen, among other types and modes of information. Such information could be provisioned by closed-weight frontier models, open-weight models, AI agents or AI scientists, and other AI-enabled workflows. Restricting access to certain subsets of information through both model-level safeguards and controlled-access platforms represents one way to prevent potential misuse. We identified the three mitigations within this mitigation class as the highest-priority efforts for industry, government, and civil society stakeholders to pursue. This is because of the mitigations' combined ability to close off common information access points that provide the foundation for real-world capability deployment: open-weight and closed-weight GPAI systems and AI-enabled BTs.

Development of Tamper-Resistant Safeguards for Open-Weight AI Models

Problem Overview

Open-weight models with high biological capabilities, including both large language models (LLMs) and AI-enabled BTs, present a critical challenge for restricting access to dangerous and dual-use information—once the model's weights are released, a model developer cannot easily enforce centralized or robust safeguards to prevent misuse. Although actors can jailbreak both closed-weight and open-weight models, open-weight models pose additional risks. For example, actors can remove safeguards at the model-weight level (e.g., an abliterated model) or retrain an open-weight model with a specific dataset to increase the model's misuse potential (e.g., a fine-tuned model). Open-weight models from both the United States and China—both abliterated and fine-tuned varieties—have recently been shown to lag closed-weight counterparts by six to 12 months on certain capability sets (the shorter estimates are based mostly on public benchmark performance).[2] Although some assessments have disputed the exact lag time and robustness of measured capabilities,[3] open-weight models present a potentially unregulated source of information for nefarious actors. This is currently an unmanageable risk because there are no known scalable ways to robustly safeguard open-weight models against misuse once released.

How This Mitigation Reduces Risks

This mitigation aims to invest in the technical development of novel and scalable approaches for eroding base capabilities with a high potential for misuse. This could prevent potential adversarial fine-tuning of AI models to hazardous levels while maintaining the models' beneficial capabilities. Such investment by government or industry would aim to definitively determine whether these technical safeguards are sufficiently

[2] Ben Cottier, Josh You, Natalia Martemianova, and David Owen, "How Far Behind Are Open Models?" EpochAI, November 4, 2024; Nestor Maslej, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Njenga Kariuki, Emily Capstick, Anka Reuel, Erik Brynjolfsson, John Etchemendy, et al., *The AI Index 2025 Annual Report*, AI Index Steering Committee, Institute for Human-Centered AI, Stanford University, April 2025.

[3] Center for AI Standards and Innovation, *Evaluation of DeepSeek AI Models*, National Institute of Standards and Technology, 2025; National Institute of Standards and Technology, "CAISI Evaluation of DeepSeek AI Models Finds Shortcomings and Risks," press release, last updated November 20, 2025.

robust at scale. Investment would also engage the open-weight model community in urgently implementing the mitigation across relevant GPAI systems and AI-enabled BTs.

For this mitigation to be effective, we assume the following:

- Open-weight GPAI models will continue to advance in biological capability and will eventually approach or match the uplift risk that frontier closed-weight models pose.
- Open-weight AI-enabled BTs with sufficient accuracy and efficiency will be required to adequately design an enhanced or novel biological threat that could incite a high-consequence biological incident.
- The complete restriction of open-weight model releases above certain capability or size thresholds is unlikely without binding government regulation.
- Technical safeguards are a necessary complement to any policy-based approach and are only viable as a mitigation if they preserve a model's utility for legitimate and beneficial use cases, which makes community engagement and adoption of the mitigation a necessary condition for implementation at scale.

Mitigation Requirements and Limitations

Existing approaches to open-weight model safety have demonstrated significant limitations for both LLMs and BTs. *Unlearning*, a technique that applies posttraining algorithms to suppress specific capabilities using example datasets, has proven insufficiently robust. Models that have undergone unlearning can readily reacquire suppressed capabilities when re-exposed to relevant data during fine-tuning.[4] This makes posttraining suppression inadequate against actors who possess appropriate datasets and the technical expertise to retrain a model. More-recent research has demonstrated that safeguards are substantially more resistant to adversarial circumvention when implemented prior to or during base-model training rather than applied afterward.[5] This approach, sometimes characterized as instilling deep ignorance into the model's weights, has shown preliminary resistance to adversarial fine-tuning in initial studies, although the approach's robustness has not yet been validated at the scale of frontier models. Critically, for such safeguards to be acceptable to the open-weight model community and to avoid foreclosing beneficial applications, the safeguards must be designed to degrade only specific capabilities while fully preserving performance on legitimate scientific, medical, and research tasks.

These efforts become more urgent as open-weight models continue to progress in capability, potentially reaching capabilities at the frontier of closed-weight models. Developing a frontier of comparable tamper-resistant safeguards is vital for the AI ecosystem to determine the acceptable risks of a potential model weights release.

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[4] Stephen Casper, Kyle O'Brien, Shayne Longpre, Elizabeth Seger, Kevin Klyman, Rishi Bommasani, Aniruddha Nrusimha, Ilia Shumailov, Sören Mindermann, Steven Basart, et al., "Open Technical Problems in Open-Weight AI Model Risk Management," *Transactions on Machine Learning Research*, March 2026; Boyi Wei, Zora Che, Nathaniel Li, Udari Madhushani Sehwag, Jasper Götting, Samira Nedungadi, Julian Michael, Summer Yue, Dan Hendrycks, Peter Henderson, et al., "Best Practices for Biorisk Evaluations on Open-Weight Bio-Foundation Models," arXiv, arXiv:2510.27629, October 31, 2025; James R. M. Black, Moritz S. Hanke, Aaron Maiwald, Tina Hernandez-Boussard, Oliver M. Crook, and Jaspreet Pannu, "Open-Weight Genome Language Model Safeguards: Assessing Robustness via Adversarial Fine-Tuning," arXiv, arXiv:2511.19299, November 24, 2025.

[5] Kyle O'Brien, Stephen Casper, Quentin Anthony, Tomek Korbak, Robert Kirk, Xander Davies, Ishan Mishra, Geoffrey Irving, Yarin Gal, and Stella Biderman, "Deep Ignorance: Filtering Pretraining Data Builds Tamper-Resistant Safeguards into Open-Weight LLMs," arXiv, arXiv:2508.06601, August 8, 2025; Yadan Chen, Mycal Tucker, Nina Panickssery, Tony Wang, Francesco Mosconi, Anjali Gopal, Carson Denison, Linda Petrini, Jan Leike, Ethan Perez, and Mrinank Sharma, "Enhancing Model Safety Through Pretraining Data Filtering," *Alignment Science Blog*, August 19, 2025; Igor Shilov, Alex Cloud, Aryo Pradipta Gema, Jacob Goldman-Wetzler, Nina Panickssery, Henry Sleight, Erik Jones, and Cem Anil, "Beyond Data Filtering: Knowledge Localization for Capability Removal in LLMs," arXiv, arXiv:2512.05648, December 5, 2025.

This mitigation is directly strengthened by the companion mitigation on specialized biological data protection. Restricting access to the laboratory-validated datasets that are most relevant to pathogen design raises the cost of adversarial fine-tuning that would circumvent even robust pretraining safeguards. The primary limitations of this mitigation class are the unresolved questions about the scalability and robustness of existing pretraining approaches, the voluntary nature of adoption absent any regulation, and the inability of any safeguard implemented by U.S.-based developers to constrain the release of open-weight models by developers operating in other nations.

Protection of Sensitive Biological Data

Problem Overview

Access to laboratory-validated, high-risk biological datasets could enable a nefarious actor using AI tools to train, develop, or fine-tune an AI model to accelerate the design or optimization of pathogens. The risk is that such data can be used to improve the biological capability of an existing AI model through fine-tuning, or the data can even be the foundation for training an entirely new tool with targeted biological capabilities. Access to the right amount, quality, and type of data could be a force multiplier to enable the execution of dangerous tasks, such as the design and enhancement of eukaryotic viruses through the training, development, or refinement of powerful AI tools and platforms.[6] Such datasets might include libraries of viral variants that can be used to enhance transmissibility, virulence, or immune evasion compared with a virus's wild type,[7] although more research is needed to clearly define the quantity and quality of data that present the highest risks.

How This Mitigation Reduces Risks

By restricting access to high-risk biological data using stricter publication norms, differential access platforms, or centralized secure repositories, responsible actors across the research ecosystem can deny potential adversaries the fuel required for adversarial fine-tuning of GPAI models or AI-enabled BTs.[8] Furthermore, restricting access to raw biological data might be a stronger protection against future misuse than restricting access to tools themselves because early research indicates that coding agents might be able to train and develop such tools when given access to the appropriate data and context for the tool.[9] Research organizations are also actively developing ways for tool development to become more automatable for beneficial research purposes, further indicating that restrictions of the tools themselves may not be sufficiently robust or acceptable to the research community.[10]

For this mitigation to be effective, we assume the following:

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[6] Allison Berke, Forrest W. Crawford, Toby Webster, James Smith, Sana Zakaria, and Sella Nevo, *Data and AI-Enabled Biological Design: Risks Related to Biological Training Data and Opportunities for Governance*, RAND Corporation, PE-A3886-1, June 2025.

[7] Pannu, Bloomfield, et al., 2025.

[8] Doni Bloomfield, James R. M. Black, Oliver Crook, Nadav Brandes, Moritz S. Hanke, Thomas V. Inglesby, Anita Cicero, Robert Pollack, Tina Hernandez-Boussard, Michael J. Imperiale, et al., "Biological Data Governance in an Age of AI," *Science*, Vol. 391, No. 6785, February 5, 2026.

[9] Luca Righetti, Kamile Lukosiute, and James Black, "Coding Agents Are Changing the Biosecurity Risk Landscape," *GovAI* blog, April 20, 2026.

[10] Biomni, homepage, undated.

- Certain specialized laboratory-validated datasets could meaningfully uplift biological weapon design capabilities when used to fine-tune or improve GPAI models or AI-enabled BTs beyond what is achievable with existing publicly available data.
- AI-enabled BTs are not yet powerful enough—when trained exclusively on publicly available data—to reach the most-dangerous capability thresholds, creating a time-sensitive window in which protecting the most-relevant datasets can materially slow the development of increasingly capable and dangerous models.
- Laboratory-validated biological data have significantly more utility over synthetically generated biological data for training a model with efficient and accurate biodesign capabilities.

Mitigation Requirements and Limitations

For this mitigation to function as a meaningful barrier, consensus must first be established on what categories and quantities of biological data constitute a meaningful fine-tuning or training risk. Not all sensitive biological data carries equal risk, and an overly broad definition would impose unnecessary costs on beneficial research, while an overly narrow definition would leave important gaps. Once the scope of high-risk data is defined, the implementation architecture must address both access and security. A universal credentialing and managed-access system, one possibly analogous to those developed for controlled human genomic data, would allow legitimate researchers to continue accessing necessary datasets while excluding actors who cannot demonstrate appropriate authorizations and purposes. A distinct challenge is the potential tension between data-protection norms and long-standing commitments to openness in fundamental research, including the principles established under National Security Decision Directive 189,[11] which has historically been interpreted as favoring broad access to the outputs of federally funded basic research. Navigating this tension will require deliberate engagement with the research community and a carefully scoped regulatory rationale that distinguishes data protection for biosecurity purposes from broader restrictions on scientific communication. Precedents from human genomic data governance, in which privacy and security concerns have justified managed-access frameworks without foreclosing research, offer a useful model for both the rationale and the mechanics of implementation.

Key features of this mitigation class include (1) developing consensus on the quantity and types of biological datasets that could enable malicious fine-tuning, (2) establishing norms that such data should be closely managed by responsible actors across the ecosystem, (3) implementing a universal system of credentialing and managed access to support responsible innovation, and (4) deploying appropriate cybersecurity controls to prevent unauthorized access or data exfiltration. Importantly, this mitigation class will no longer be relevant should the value of laboratory-validated biological data be eclipsed by in silico and virtual research methods that are capable of producing synthetic datasets of equivalent utility. Leading indicators of such a shift should be identified, tracked closely, and used to trigger a reassessment of this mitigation's prioritization and implementation.

The U.S. government is central to making this mitigation class functional at the required scale and consistency. The consensus of the scientific community alone is unlikely to produce the standardized and universal adoption that is necessary for data protection to serve as a reliable barrier. Government incentives, and potentially regulatory requirements, will be needed to drive a consistent implementation across academic institutions, private research organizations, and international partners. Federal agencies with equities in biological research funding are well positioned to attach data-governance requirements to grant programs and research agreements as an initial lever. International coordination is also essential. This is because data pro-

[11] Office of the President of the United States, *National Policy on the Transfer of Scientific, Technical and Engineering Information*, NSDD-189, September 21, 1985.

tection efforts that apply only within the United States create opportunities for international collaborators or foreign institutions operating under different norms to generate, store, or access high-risk datasets. Engagement through multilateral scientific governance bodies and bilateral research agreements will be needed to extend data-protection standards beyond the United States.

Denying access to sensitive biological data could enforce barriers for less capable actors, including lone actors and some nonstate groups that might seek to design or develop an enhanced or novel biological weapon. Although data protection cannot fully stop insider threats or state programs that are capable of generating laboratory-validated data independently, it might slow the threat progression for more opportunistic actors and raise the cost of AI-enabled biological weapons development across a broad variety of threat-actor profiles.

Access-Control Platforms and Mechanisms for Dual-Use AI Systems

Problem Overview

Prior to the emergence of GPAI, a nefarious actor seeking to develop biological weapons would rely primarily on internet searches to access dangerous information; searches could be of declassified records of historical biological weapons programs or dual-use findings embedded in academic literature. AI systems have fundamentally altered this landscape by enabling more-efficient synthesis and personalization of such information and by demonstrating an emergent capacity to extrapolate novel hypotheses and approaches beyond what is explicitly represented in training data.[12] Assessment studies and benchmarks have increasingly demonstrated that AI models have reached expert-level proficiency across virology and molecular biology tasks,[13] effectively placing the functional equivalent of a specialized scientific consultant within reach of any actor with AI model access. Openly available models, including open-weight GPAI systems and AI-enabled BTs, provide this access with limited mechanisms for monitoring or disrupting potential misuse. At the same time, these capabilities provide genuine value to legitimate scientific researchers, including in the development of threat-characterization pipelines, early-warning systems, and medical countermeasures (MCMs).

How This Mitigation Reduces Risks

Access to models and tools with powerful biological capabilities must be structured in ways that preserve their beneficial uses while reducing the probability and consequence of misuse. A world that is both innovation-compatible and protected against unacceptable risk will require managed-access programs and platforms that enable responsible actors to use the tools, information, and data necessary for beneficial purposes while introducing meaningful friction for misuse. Such programs require tiered approaches and verified-user credentialing rather than default or expansive access to sensitive capabilities, and frontier AI model developers

[12] Ludovico Mitchener, Angela Yiu, Benjamin Chang, Mathieu Bourdenx, Tyler Nadolski, Arvis Sulovari, Eric C. Landsness, Daniel L. Barabasi, Siddharth Narayanan, Nicky Evans, et al., "Kosmos: An AI Scientist for Autonomous Discovery," arXiv, arXiv:2511.02824, November 4, 2025.

[13] Jasper Götting, Pedro Medeiros, Jon G. Sanders, Nathaniel Li, Long Phan, Karam Elabd, Lennart Justen, Dan Hendrycks, and Seth Donoughe, "Virology Capabilities Test (VCT): A Multimodal Virology Q&A Benchmark," arXiv, arXiv:2504.16137v2, April 29, 2025; Chen Bo, Calvin Zhang, Christina Q. Knight, Nicholas Kruus, Jason Hausenloy, Pedro Medeiros, Nathaniel Li, Aiden Kim, Yury Orlovskiy, Coleman Breen, et al., "LLM Novice Uplift on Dual-Use, In Silico Biology Tasks," arXiv, arXiv:2602.23329, February 26, 2026; Sunishchal Dev, Charles Teague, Grant Ellison, Kyle Brady, Ying-Chiang Jeffrey Lee, Sarah L. Gebauer, Henry Alexander Bradley, Dawid Maciorowski, Bria Persaud, Jordan Despanie, Barbara Del Castello, Alyssa Worland, Michael Miller, Adrian Salas, Dave Nguyen, James Liu, Jason Johnson, Andrew Sloan, Will Stonehouse, Travis Merrill, Thomas Goode, Greg McKelvey, Jr., and Ella Guest, *Toward Comprehensive Benchmarking of the Biological Knowledge of Frontier Large Language Models*, RAND Corporation, RR-A3797-1, 2025; Brady et al., 2026.

could implement these approaches through specialized programs,[14] or they can do so through downstream deployment contexts in which models and tools are scaffolded and/or fine-tuned for particular purposes.[15] Permissions can be structured across several dimensions: tiered capability access that restricts the most sensitive functionalities to users who demonstrate clear justification, appropriate training, and alignment with defined research objectives; time-limited access that reduces persistent exposure to high-risk capabilities; justified use requirements that mandate articulations of legitimate purposes before sensitive functionalities are granted; and auditability that ensures that access and use patterns can be reviewed if concerns arise. These controls do not eliminate risk but rather introduce friction that deters opportunistic misuse, and the controls could generate investigative leads when activity deviates from expected patterns. In combination with technical safeguards and active monitoring, scope-bound permissions strengthen both deterrence and disruption pathways.[16]

Because *insider threats*, which we define as actors who are credentialed members of the legitimate scientific community, represent a genuine and distinct risk category, managed-access platforms must not function as a gatekeeping exercise that ends at the point of initial credentialing. Access to the highest-risk tools and datasets should carry additional provisions, including privacy-preserving, real-time monitoring of use patterns and periodic recredentialing, to surface potential misuse originating from within the authorized user population.[17] The monitoring infrastructure embedded in these platforms also has direct value for the strategy's proactive detection mitigations that we describe elsewhere in this report; the infrastructure provides the audit trail and behavioral signal that support both early-misuse identification and downstream attribution.

For this mitigation to be effective, we assume the following:

- Open-weight and closed-weight AI models and tools with expert-level biological capabilities meaningfully lower the barrier to biological weapons development for actors who would otherwise lack access to specialized expertise.
- The same capabilities are sufficiently valuable to legitimate researchers, and restricting access entirely would impose unacceptable costs on beneficial innovation.
- A tiered-access architecture that structures permissions around purpose, user identity, capability level, and context can reduce misuse risks without foreclosing legitimate use, but only if verification and monitoring requirements are enforced with adequate rigor.

Mitigation Requirements and Limitations

Engaging the U.S. government is central to this mitigation class's effectiveness. Voluntary industry and academic action has produced early examples of tiered access and know-your-customer practices among some model providers and BT developers. However, coverage is uneven and standards are inconsistent across the ecosystem. The government should establish minimum requirements for access controls, user verification, and audit logging that are applicable to AI models and BTs that are above defined risk thresholds, and the government should do so while providing clear guidance on what constitutes adequate credentialing and monitoring for different capability levels. Federal investment in shared credentialing infrastructure could

[14] OpenAI, "Strengthening Societal Resilience with Rosalind Biodefense," May 29, 2026; Anthropic, "Introducing Anthropic's AI for Science Program," press release, May 5, 2025a.

[15] CEPI, "Securing the Pandemic Preparedness Engine," September 18, 2025.

[16] Jonathan Feldman, Tal Feldman, and Annie I. Anton, "Know Your Scientist: KYC as Biosecurity Infrastructure," *Frontiers in Microbiology*, Vol. 17, April 29, 2026; Sarah R. Carter and Greg Butchello, *A Framework for Managed Access to Biological AI Tools*, Nuclear Threat Initiative, January 28, 2026.

[17] Sarah R. Carter, *Developing a Customer Screening Framework for the Life Sciences*, Blueprint Biosecurity, March 2024.

reduce the burden on individual providers and accelerate the consistent adoption of standards across the ecosystem.

Key features of this mitigation class include (1) a tiered credentialing architecture that matches access permissions to demonstrated purposes and qualifications, (2) time-limited and scope-bound access provisions that reduce persistent risk exposure, (3) audit and monitoring infrastructure that is capable of detecting anomalous use by both external actors and credentialed insiders, (4) and standards and reporting protocols that connect these platforms to broader detection and response frameworks.

This set of mitigations is feasible to implement through concerted industry and academic efforts, but it is unlikely to significantly reduce risk in isolation because there are threat actors for whom information access does not represent a primary barrier. For those actors, the deterrence and physical chokepoint interventions that we describe in other sections of this report will be more effective, and the full value of access-control platforms is realized only in combination with those complementary measures.

Disrupt Materials Access

There are some threat actors who already have the expertise and resources necessary to effectively design and develop a pathogen, plan and execute successful attack, and/or circumvent biosecurity measures. They are able to do so because of protection and resourcing from state governments, or they have unique access because they already have legitimate credentials or experience; for example, they are a scientist or researcher. It is challenging to disrupt such actors, and it is likely that they would be unaffected by mitigations within this category. However, actors who have limited resources or are acting alone are more likely to seek access to materials that are available on the commercial market, setting up an opportunity for a potential disruption of biological misuse.

Biological Precursor Screening

Problem Overview

The design and development of a biological weapon rely on an actor's ability to access the raw materials necessary for synthesizing, manipulating, or producing a pathogen in the laboratory. Access to nucleic acids, pathogen or toxin laboratory stocks, or other precursor components are considered an important bottleneck in a nefarious actor's ability to develop a functional biological agent. This is because the precision and scale of synthesis required for weaponization exceeds what some actors could achieve through purely in-house or informal means.

How This Mitigation Reduces Risks

In addition to strengthening and mandating synthesis screening, there is a need to implement similar monitoring and screening of other precursors, including those that come from providers outside the gene-synthesis industry. Examples of such providers are cloud lab providers, contract research organizations, and plasmid repositories. Biological-product screening and *know-your-customer screening*—the practice of evaluating orders for dangerous sequences or components and flagging suspicious customer behavior before fulfillment—represent some of the most operationally concrete and grounded mitigations available to interrupt the biological risk chain at a physical chokepoint that AI tools cannot themselves eliminate. Over the past two decades, industry-led organizations and gene-synthesis companies have driven the charge to set standards for implementing robust gene-synthesis-screening practices with a focus on nucleic acid fragments and oligonucleotides as the primary screened product.

Although the U.S. government has recently worked to complement these industry-led actions using federal policy incentives and potential regulatory frameworks,[18] the existing baseline of biological precursor screening across industry must be substantially strengthened in both coverage and quality. As of this writing in July 2026, both product screening and know-your-customer screening are uneven across providers and largely voluntary, and the most mature programs are concentrated among large commercial gene-synthesis companies. A comprehensive screening architecture must extend beyond traditional oligonucleotide and gene-synthesis suppliers to include any organization (e.g., repositories, contract research organizations, cloud laboratories) that provisions biological components relevant to pathogen construction, including viral vectors, viral preparations, select derivatives, and assembled genetic constructs.

For this mitigation to be effective, we assume the following:

- A high-consequence biological incident will primarily be possible through the design and development of a biological agent de novo,[19] which requires biological precursors.
- Obtaining biological precursors through commercial synthesis channels is a major chokepoint for actors who are not part of a state's biological weapons program or the legitimate scientific community, and they are unlikely to possess the equipment, expertise, and supply chains necessary to independently produce sufficiently accurate constructs.
- A strengthened and coordinated precursor-screening ecosystem can meaningfully raise the barrier to biological weapons development even as AI tools lower barriers to their design.
- The coordination and reporting infrastructure required for effective screening creates a unique and direct interface with law enforcement, enabling intervention before an actor has advanced to the point of developing a biological weapon.

Mitigation Requirements and Limitations

As with the protection of biological data, engagement with the U.S. government is essential to realizing this mitigation. Voluntary industry programs have established proofs of concept, but achieving the necessary coverage, consistency, and coordination for screening to function as a genuine deterrent requires a regulatory foundation. Furthermore, the precursor products and providers that implement both product screening and know-your-customer screening must expand to include not just gene-synthesis providers but also existing and emerging industries, such as contract research organizations, cloud laboratories, and plasmid repositories. The government should establish mandatory screening requirements with defined minimum standards for precursor screening, customer vetting, and data-sharing and that are applicable across provider types and enforced with sufficient oversight to prevent regulatory arbitrage. Federal investment in shared threat-database infrastructure and screening-algorithm development would reduce the cost burden on individual providers and accelerate the adoption of next-generation functional screening methods. Critically, the reporting and coordination infrastructure built into screening programs must be directly connected to law enforcement and intelligence community channels so that flagged orders and suspicious patterns generate actionable investigative leads rather than simply resulting in a declined transaction.

On the technical side, product-screening methodologies must also evolve to remain effective against AI-enabled threat pathways. Existing screening approaches rely heavily on sequence homology to known list-

[18] See Executive Order 14292, "Improving the Safety and Security of Biological Research," Executive Office of the President, May 5, 2025; and U.S. Department of Health and Human Services, Administration for Strategic Preparedness and Response, *Screening Framework Guidance for Providers and Users of Synthetic Nucleic Acids*, October 2023.

[19] In this case, *de novo* refers to how actors will need to assemble a biological agent from such precursor components as deoxyribonucleic acid (DNA), ribonucleic acid (RNA), or other nucleic acids rather than obtaining the sample from nature or a laboratory stock.

based dangerous pathogens and select pathogens and toxins, but AI tools are increasingly capable of designing novel sequences that retain dangerous functions while diverging sufficiently from known references to evade homology-based detection.[20] Therefore, ongoing investment in the screening algorithms is required, including making regular updates to address vulnerabilities that adversarial design techniques—potentially AI-assisted—might exploit. Finally, the proliferation of benchtop DNA synthesizers presents a parallel challenge: As synthesis capability becomes more distributed and more accessible, screening must be embedded within the synthesizer hardware and software so that the physical instrument enforces screening requirements regardless of the commercial channel through which it is used.

Synthesis screening is among the most frequently cited and deeply analyzed mitigations in biosecurity policy discussions, and there is a substantial bench of technical and policy expertise available to inform its implementation. This is likely because the field of synthesis screening originated in the nucleic acid provider community and represents a truly industry-led biosecurity measure over almost the past two decades. Synthesis screening's primary limitations are the coverage gaps that persist until all relevant provider types are included, the ongoing adversarial dynamic between screening methods and AI-assisted design of evasive sequences, and the fact that state biological weapons programs with indigenous synthesis capabilities operate entirely outside the commercial channels that this mitigation targets. Common know-your-customer standards and coordinated data-sharing protocols across providers would enable the detection of *split orders*—that is, a nefarious actor divides a dangerous synthesis request across multiple vendors to evade any single provider's screening threshold—as well as patterns of escalating or anomalous procurement behavior that are only visible in aggregate.

Key features of this mitigation class include universal product-screening and know-your-customer screening coverage across all relevant provider types, including an expansion to repositories, cloud and automated laboratories, and contract research organizations; shared and standardized threat databases and know-your-customer criteria; international coordination that is sufficient to eliminate jurisdictional loopholes; functional inference-screening capabilities that are resilient to AI-assisted sequence obfuscation; the integration of screening into benchtop synthesis hardware; and law enforcement coordination protocols that convert screening flags into investigative action.

Although this mitigation is only robust against threat actors who do not have alternative or legitimate mechanisms to obtain biological precursors (e.g., a member of the scientific community or an actor embedded in a state's biological weapons program), it can meaningfully add friction to nonstate actors without these access privileges.

Deter Nefarious Activity

Not all actors who might seek to exploit AI-enabled biological capabilities are meaningfully constrained by access barriers. A sufficiently motivated actor, whether a state-sponsored program, a state-adjacent proxy, or a nonstate group with adequate resources and insider access, might be able to circumvent information controls, source sensitive data independently, or develop a dangerous capability through channels that mitigation efforts—particularly efforts targeting model access and data protection—cannot fully reach. For these actors, preventing their nefarious activity must occur through a different mechanism, specifically through deterrence that is achieved by raising the perceived cost of action and/or degrading the expected utility of an

[20] Bruce Wittmann, Tessa Alexanian, Craig Bartling, Jacob Beal, Adam Clore, James Diggans, Kevin Flyangolts, Bryan T. Gemler, Tom Mitchell, Steven T. Murphy, et al., "Strengthening Nucleic Acid Biosecurity Screening Against Generative Protein Design Tools," *Science*, Vol. 390, No. 6768, October 2, 2025.

attack. Two key conditions that shape whether a prospective actor is deterred are their belief that they will be identified and held accountable and their belief that their attack will not achieve its intended effect. The three mitigations within this mitigation class target each of these conditions directly: Bioattribution and biological forensic capabilities increase the credible risk of detection and consequences for actors who care about exposure, pathogen early-warning systems compress detection timelines in ways that reduce an attacker's confidence in achieving mass impact, and a demonstrated response capability reduces the expected attack utility by showing that response measures can be developed and deployed quickly, including against novel agents. A foundation for the deterrent effect of these mitigations is that the capability is both known and credible, meaning that the demonstration of performance through regular exercises and dedicated investments is critical.

Together, the mitigations represent the highest-priority deterrence investments for government and its partners in the near term, and we selected them for their ability to extend deterrence coverage to threat actors and scenarios that access-control-oriented mitigations alone cannot address.

Bioattribution and Forensics

Problem Overview

Effective deterrence through the possibility of consequences requires that prospective threat actors believe that they will be identified and held accountable for harmful acts. *Bioattribution*—the capacity to credibly link a biological attack or the misuse of AI tools for biological weapons development to a specific actor—is therefore a foundational component of any deterrence-based mitigation strategy.

How This Mitigation Reduces Risks

Although some threat actors might be largely indifferent to exposure (e.g., certain lone actors in an acute crisis), a meaningful subset of concerning actors—including state-sponsored programs, state-adjacent or proxy actors, ideologically motivated nonstate groups with reputational or political stakes, and criminal enterprises—do care about getting caught. For these actors, a credible and publicly known attribution architecture can raise the perceived cost of action, introduce hesitation, and strengthen the deterrence posture that complements more access-oriented mitigations, such as data protection or model-access controls. Such attribution capability must be able to achieve a timely and high-confidence assessment to lead to action at the speed of relevance—that is, this capability should be regularly exercised on outbreaks around the world.

For this mitigation to be effective, we assume the following:

- Some fraction of actors who might seek to conduct an AI-enabled biological attack are deterred by the prospect of attribution and downstream consequences, including geopolitical, legal, or reputational costs.
- Attribution capability, even if imperfect, functions as a deterrent when threat actors believe that the capability is sufficiently mature and will be exercised.
- Attribution in this domain requires the integration of both biological forensic methods and digital forensic methods given the nature of AI-enabled threats that span both digital and physical signatures along the design and development processes.

Mitigation Requirements and Limitations

Attribution in this context encompasses two distinct but related technical tracks. The first is biological forensic attribution, including microbial-forensic capabilities that can characterize the genetic, phenotypic, and production signatures of a weaponized agent to identify its provenance, likely engineering history, and possible point of origin. AI-assisted tools may become increasingly capable of reverse-engineering pathogen-

modification strategies from sequencing data, which could help analysts identify whether and how AI tools were used in weapon design—potentially linking an agent's design to a specific model architecture, training corpus, or fine-tuning dataset. The second track is digital forensic attribution, leveraging the audit trails, access logs, and use metadata generated by the access-control platforms and mechanisms described previously. Queries to AI-enabled BTs, data-access events, model outputs, and deviations from expected use patterns can generate investigative leads that, when aggregated and analyzed, might help attribute misuse to specific individuals, organizations, or nation-state programs. The watermarking of AI-generated biological content—an emerging technical area—could further strengthen the ability to trace outputs to their originating model or session. These two tracks are mutually reinforcing: Digital forensics can constrain the field of suspects, while biological forensics can confirm or exclude candidate attributions.

The technical development of this mitigation class is nascent and requires deliberate investment. Microbial-forensics capabilities, while developed in the context of biodefense and law enforcement following the anthrax events that occurred after the September 11, 2001, terrorist attacks, have not been systematically updated to address AI-enabled threat pathways. Benchmarks and validated methods for attributing AI-assisted pathogen design do not yet exist and represent a significant gap. On the digital side, the attribution value of access-control and monitoring infrastructure directly depends on the quality of logging standards, data-retention policies, and interoperability between platforms—areas that require coordination between government and industry.

Critically, the deterrent value of attribution is not solely a function of technical capability but of U.S. posture and signaling. A government that possesses attribution capabilities but does not communicate them credibly or exercise them with visible consequences will not achieve deterrence. This mitigation class therefore requires an explicit policy component: The U.S. government should develop and publicly communicate a doctrine of consequence for AI-enabled biological threats, including clear statements that attribution will be pursued and that credible attributions will trigger defined responses across diplomatic, legal, economic, and, potentially, military channels. Investment in attribution capabilities should be paired with investment in the consequence frameworks that give attribution meaning—including international legal instruments, export-control enforcement, sanctions authorities, and law enforcement channels that are capable of acting on digital forensic leads. International cooperation on attribution standards and evidence-sharing agreements, such as agreements that are analogous to those developed for cyberincident attribution, will be essential to extend deterrence to actors operating across jurisdictions.

Key features of this mitigation class include (1) the development of validated forensic methods for AI-assisted biological threat attribution, (2) the integration of attribution infrastructure into access-control and monitoring platforms from the design stage, (3) U.S. government doctrine and signaling that communicates attribution intent and capability, and (4) consequence frameworks that are capable of translating attribution into meaningful costs. This mitigation class is unlikely to achieve its deterrent potential without deliberate government investment and a credible track record of attribution and response. Highly sophisticated state actors with indigenous AI and biological capabilities might operate in ways that evade attribution, and some actors remain undeterred regardless of their exposure to risk. For those actors, attribution-based deterrence must be combined with the denial-oriented mitigations that we describe elsewhere in this report.

Pathogen Early-Warning Systems

Problem Overview

A threat actor weighing whether to deploy a biological weapon must assess the risk of facing consequences for their actions. A passive, continuously operating, and pathogen-agnostic early-warning system—one that does not depend on a clinical recognition of disease, physician reporting, or population-level cooperation to

function—could meaningfully compress the window between release and detection, improving the speed and confidence of attribution. As discussed in the previous section, confident and credible attribution makes successful punitive action against the perpetrator more likely, thereby strengthening deterrence by consequence for a variety of actors, including those not easily detected or disrupted (e.g., a state's biological weapons programs).

How This Mitigation Reduces Risks

Implementing and regularly exercising a highly capable early-pathogen detection system, in combination with attribution capabilities, could deter actors from opting to create a biological weapon in the first place because of concerns that their activities could be detected early and attributed with confidence.

For this mitigation to be effective, we assume the following:

- Some fraction of actors who might consider deploying an AI-enabled biological agent are deterred when the probability of consequence and punishment is perceived to be high (e.g., deterrence by consequence).
- A passive environmental surveillance system across multiple modalities that operates continuously and without requiring symptomatic individuals to seek care provides meaningfully earlier warning than clinical or syndromic surveillance alone.
- Early-warning systems can generate valuable forensic signals that support downstream attribution.

Mitigation Requirements and Limitations

Wastewater epidemiology has emerged as an operationally mature candidate for a passive, pathogen-agnostic early-warning platform.[21] The COVID-19 pandemic demonstrated that wastewater surveillance can detect pathogen circulation in communities days to weeks before clinical case counts rise, and it can do so at a fraction of the cost of active surveillance programs. Critically, wastewater-based surveillance does not require any action or cooperation from the affected population—it operates on biological signals that are shed passively and continuously, integrating exposure across large catchment areas. Advances in metagenomic sequencing might enable the detection and characterization of novel or engineered pathogens without requiring prior knowledge of the target organism, which is essential given that AI-enabled biological threats might involve agents specifically designed to evade known diagnostic panels. A sufficiently dense and sensitive national wastewater surveillance network, combined with the use of routine metagenomic sequencing and anomaly-detection algorithms through other channels—such as routine surveillance of the health care system and a targeted deployment of wearable technology to high-risk populations—could provide early signals of both natural spillover events and deliberate release without presupposing what the threat agent will be. The data that such a system generates also have direct forensic value: Genomic signatures that are captured in environmental samples at the earliest stages of an incident can inform microbial forensic attribution efforts, potentially preserving evidence that would otherwise be lost as an outbreak scales up and the epidemiological picture becomes more complex.

The deterrent value of this architecture depends, in part, on the architecture being known and credible. A system that exists but is not widely understood to exist provides a limited deterrence signal. The U.S. government should therefore include deliberate public communication about the maturity and coverage of environmental surveillance capabilities, and it should regularly exercise the capability through the normal

[21] We focus on wastewater surveillance here as an example of how pathogen early-warning systems provide value. But other passive biosurveillance modalities, such as air sampling, surveilling food and agriculture systems, or international travel surveillance data, might also have utility. See Henry H. Willis, Adeline E. Williams, Saskia Popescu, Derek Roberts, Eva Coringrato, Laura J. Faherty, Pedro Nascimento de Lima, and Sana Zakaria, "A Framework for Evaluating Implementation, Impact, and Cost-Effectiveness of Wastewater and Environmental Surveillance," *Frontiers in Public Health*, Vol. 14, March 18, 2026.

course of public health responses as part of a broader signaling strategy that raises the perceived detection risk for prospective actors.

Key features of this mitigation class include (1) infrastructure density and coverage that are sufficient to eliminate surveillance gaps, (2) analytical pipelines that are capable of pathogen-agnostic anomaly detection in metagenomic data, (3) integration with forensic and attribution workflows to preserve early-stage genomic evidence, and (4) the U.S. government's signaling posture that communicates surveillance maturity to prospective actors. This mitigation class provides compounding returns: The same infrastructure strengthens both the attribution capability and public health response, making it among the higher-leverage investments in a layered mitigation strategy. Its principal limitations are the significant infrastructure investment required for national-scale coverage, the resources needed to regularly exercise and communicate about the capability even in times of a nonpandemic crisis, and the possibility that a sufficiently targeted or localized release could fall below detection thresholds in sparse sampling networks.

Demonstrated Rapid-Response and Consequence-Mitigation Capabilities

Problem Overview

A threat actor weighing whether to develop or deploy a biological weapon must assess not only whether they are likely to be detected or attributed but also whether the effects they seek to generate can be meaningfully blunted on timelines that matter for strategic impact. Rapid-response consequence-mitigation capabilities should be understood not only as downstream response tools but also as components of deterrence by denial when they are sufficiently adaptable, scalable, and operationally demonstrated to remain effective under uncertainty. These capabilities might include MCMs, diagnostics, personal protective equipment (PPE), environmental decontamination measures, indoor-air interventions, health care surge capacity, nonpharmaceutical interventions, and any future consequence-mitigation technologies that might emerge as biological threats and defensive capabilities evolve. MCM platforms are a useful illustrative example, but the core deterrent value lies in demonstrating the ability to rapidly identify threats and deploy effective mitigations at meaningful scales.

How This Mitigation Reduces Risks

A system that has shown that it can repeatedly reduce the consequences of novel, engineered, or otherwise unexpected biological threats decreases an attacker's confidence that an attack will achieve durable effects at the intended scale, thus changing the expected utility of attack. The central values of such a capability are credibility and speed. A scientifically promising platform or technology that has not been operationally exercised does little to alter an adversary's assessment of likely success because theoretical promise is not the same thing as demonstrated performance under real constraints. By contrast, a standing rapid-response architecture that has demonstrated the ability to respond to any outbreak—natural, accidental, or deliberate—quickly across the full response chain (from threat identification to the deployment of effective mitigations) sends a consequential signal. Demonstrated performance across a broader response chain provides evidence that biological effects can be reduced before they achieve their intended strategic impact.

Furthermore, AI-enabled offensive capabilities might allow the use of modified, synthetic, or optimized biological pathogens and toxins that are specifically designed to exploit known gaps in existing defenses. In such an environment, deterrence increasingly depends on adaptable, capability-based approaches that can respond effectively even when the threat's specific characteristics are uncertain. Platform-based MCM technologies remain particularly valuable because they can be adapted across pathogen classes and deployed at scale, but they represent only one component of a broader rapid-response and consequence-mitigation architecture. The objective should be to demonstrate a credible capacity to rapidly reduce the consequences

of a biological attack through a combination of mitigating response measures, including the use of pathogen-agnostic platform technologies.

This mitigation represents a form of deterrence by denial that complements the deterrence-by-consequence approaches described in the previous sections. Rather than seeking to raise the costs of an attack through detection, attribution, disruption, or retaliation, the purpose of this mitigation is to reduce an actor's confidence that they can achieve meaningful biological effects in the first place. This approach might be particularly valuable for highly capable actors who are difficult to detect or disrupt before an attack occurs, including highly resourced nonstate groups and some state actors, as well as actors whose motivations make them less susceptible to threats of punishment alone. By demonstrating the ability to rapidly reduce consequences, this approach shifts the offensive calculus by reducing the expected payoff of an attack.

For this mitigation to be effective, we assume the following:

- Some actors who consider the use of AI-enabled biological pathogens are likely to be deterred when they believe that effective countermeasures can, in fact, be developed and deployed quickly enough to substantially reduce their attack's impact.
- The deterrent value of rapid-response capabilities depends meaningfully on demonstrated operational performance rather than on theoretical scientific promise.
- Adaptable response capabilities, including but not limited to platform-based MCM technologies and PPE, provide meaningful value when they can be adapted rapidly for the use against a variety of novel, engineered, or otherwise unexpected biological threats.

Mitigation Requirements and Limitations

Taken together, these assumptions support investment in a demonstrated rapid-response and consequence-mitigation architecture rather than in any single technology area alone. Investment priorities should therefore center on capabilities that strengthen demonstrated end-to-end performance, including adaptable MCM platforms that can be reconfigured across pathogen classes, rapid diagnostics, PPE stockpiles and distribution systems, indoor-air and environmental-mitigation capabilities, manufacturing systems that are capable of rapid and scalable production, regulatory and clinical mechanisms designed for an emergency evaluation of novel threats, logistics systems that support fast delivery and administration, and recurring exercises that test the full architecture under realistic conditions. Demonstration should be treated as a core preparedness output rather than a supplementary activity because public evidence of exercised readiness is what makes the capability strategically legible and credible. Rapid-response demonstrations, prototype-to-manufacture drills, demonstrated surge production, health care response exercises, and prior instances in which mitigation capabilities were successfully adapted and deployed on consequential timelines all contribute more to deterrent value than generic claims about innovation or future scientific promise. Many organizations regularly demonstrate this capability in real time when responding to actual outbreaks, and these organizations seek to strengthen and accelerate this capability through exercises, operational learning, and dedicated investments to platform development.[22] Such efforts should be elevated and messaged broadly to signal the real value and lives saved as a means of deterrent. For example, Rwanda's record-breaking response to a Marburg virus outbreak—implementing an emergency trial of a vaccine just ten days after the outbreak was declared—provides an important proof point for maximum deterrence.[23] Similar demonstrations and deployments involving diagnostics, PPE deployment, environmental controls, health care mobilization, and

[22] Kate Kelland, "The 100 Days Mission—How Close Are We?" CEPI, July 24, 2024.

[23] CEPI, "Preparedness in Practice: Rwanda's Record-Breaking Marburg Response," undated.

other rapid-response measures can also contribute to deterrence by showcasing exercised readiness and operational competence. In this respect, signaling should focus less on aspiration and more on proof of execution.

This mitigation class also differs in an important respect from traditional biodefense stockpiling. Conventional preparedness has been organized around known agents and scenario-specific inventories, and that approach remains valuable for familiar threats. However, it is less well suited to an AI-enabled threat environment in which a relevant pathogen or toxin might be novel, modified, or selected precisely to exploit known defensive gaps. The strategic value of platform technologies and other adaptable response capabilities lie in their ability to move preparedness from a threat-specific model to a capability-based model, particularly one in which the government possesses the technical and operational machinery necessary to reduce the consequences of a broad variety of plausible threats quickly enough to affect the offensive calculus.

However, rapid-response and consequence-mitigation capabilities face important limitations, such as threats that might outpace response timelines. The capabilities might also face real-world deployment constraints, such as manufacturing bottlenecks, logistical friction, and/or limited uptake, operational disruptions, or other implementation challenges. Consequently, these capabilities should not be treated as a stand-alone deterrence solution. Their values are greatest as part of a layered mitigation strategy in which demonstrated rapid-response readiness works alongside early warning and attribution to reduce an attack's expected payoff.

Detect Misuse Signals Proactively

Access controls and deterrence measures reduce the likelihood that a nefarious actor can acquire a dangerous capability or act with impunity, but neither mitigation guarantees protection against misuse attempts. Across both closed-weight frontier models and the broader ecosystem of AI-enabled BTs, individual actors might probe, query, or attempt to exploit AI systems in ways that fall below the threshold of immediate harm but nonetheless represent meaningful indicators of malicious intent or escalating capability development. Because no single model provider or institution has visibility into the full landscape of misuse activity, signals that might be actionable in aggregate can be missed or misinterpreted when observed in isolation. Proactive detection addresses this gap by creating the infrastructure and institutional arrangements necessary to identify misuse behavior in or near real time and before an actor has advanced far enough along the biological risk chain to cause harm. The two interventions in this mitigation class target the two most critical nodes in this detection architecture: the individual model level, in which real-time monitoring of query patterns and outputs by closed-weight model providers can surface early indicators of misuse, and the ecosystem level, in which a centralized information-sharing nexus at the intersection of AI and biology (AI x Bio) can aggregate, analyze, and act on signals across providers, platforms, and government stakeholders to enable a coordinated and timely response.

Real-Time Monitoring and Reporting of Misuse of Closed-Weight GPAI Models

Problem Overview

Frontier AI model developers have sought to understand and manage potential biological misuse risks of their general-purpose systems for several years by using a combination of capability evaluations and the subsequent development of safeguards. These efforts have been led by the developers themselves, a growing third-party evaluator ecosystem, and national governmental bodies, including the Center for AI Standards and Innovation in the United States and the AI Security Institute in the United Kingdom. Several frontier developers designated their mid-2025 generation models as potentially crossing the threshold of providing

meaningful technical and operational uplift to novice actors seeking to develop biological weapons.[24] With this designation came the development of more-robust posttraining safeguards, including classifiers, refusal behaviors, and safe-completion practices. These safeguards represent a genuine commitment to managing risk, but they face two limitations. First, given the inherent dual-use nature of biological information, they have produced errant account restrictions for legitimate researchers and reduced model utility in scientific discovery. Real-time misuse monitoring along with a managed-access approach could address this tension by combining aggressive default blocking with credentialed pathways to specialized capabilities, thus demonstrating the complementarity between these two mitigation classes in the strategy. Second, and more fundamentally, evidence indicates that existing safeguards can be circumvented with sufficient resources, time, and technical expertise.[25] We do not share these findings to argue for abandoning safeguards development, but we do argue that safeguards alone are insufficient and that complementary approaches are urgently needed. Real-time misuse monitoring represents one such complementary approach.

How This Mitigation Reduces Risks

Rather than relying solely on preventing a harmful query from being answered, misuse monitoring aims to detect patterns of potentially nefarious behavior across user sessions and generate actionable signals before an actor has advanced to a dangerous stage in the biological risk chain. Many frontier AI companies are already investing in technical approaches to detect potential misuse across domains, including the chemical, biological, radiological, nuclear, and high-yield explosives domain; the cyber domain; and the model distillation domain.[26] These investments provide a foundation that can be extended and specialized for biological misuse detection. Historically, it has been challenging to detect and discern the intent of a nefarious actor by how they intend to use biological materials and equipment. But the wealth of information in user logs from across millions and millions of sessions—the vast majority of which represent legitimate use of the systems for obtaining or providing biological information—presents an opportunity for an improved understanding of biological misuse and innovation pathways paired with the deployment of targeted safeguards.

Real-time misuse monitoring also presents several structural advantages over blocking access to biological capabilities. It does this by allowing better detection, deterrence, and overall risk reduction than blocking alone. When a query is blocked, the threat actor is alerted that their approach has triggered a safeguard, which can give the actor a chance to adjust their behavior. Monitoring inverts this because the threat actor is not alerted, does not change their behavior, and could provide more-useful information about future misuse. By contrast, blocking access risks displacing misuse to other environments in which monitoring is not possible, thereby reducing overall visibility into the malicious activity. A monitoring-based approach also lets organizations set detection thresholds more inclusively than would be acceptable for blocking because a flag does not result in immediate user-facing action. Ambiguous or borderline signals can be escalated to more-capable automated reviewers for more-detailed and nuanced analysis, or the signals can be escalated to a human subject-matter expert when needed and before any action is taken—reducing both false negatives and the cost of false positives compared with a system that has to make a binary block-allow decision in real time.

[24] It is important to acknowledge the complexity of such designations because both OpenAI and Anthropic indicated that they made their decisions from a place of precaution rather than definitive evidence that such uplift had been observed. Notably, there are very few methods to definitively indicate that a model can meaningfully help a human access, develop, or design a biological weapon because of the hazards inherent in measuring such outcomes directly. See OpenAI, "Preparing for Future AI Capabilities in Biology," June 18, 2025; and Anthropic, "Introducing Claude Sonnet 4.5," press release, September 29, 2025b.

[25] Xander Davies, Giorgi Giglemiani, Edmund Lau, Eric Winsor, Geoffrey Irving, and Yarin Gal, "Boundary Point Jailbreaking of Black-Box LLMs," arXiv, arXiv:2602.15001v2, February 18, 2026.

[26] OpenAI, 2025; Anthropic, 2025b.

Monitoring, when deployed appropriately, can be a fundamentally stronger mechanism for catching genuine misuse and reducing overall risk.

For this mitigation to be effective, we assume the following:

- Refusal safeguards from frontier closed-weight models will remain susceptible to circumvention by sufficiently resourced and motivated actors, making detection a necessary complement to prevention.
- The infrastructure for real-time misuse detection substantially overlaps with the infrastructure used for running the classifiers and probes used in blocking. This makes misuse detection a technically tractable area that, with focused investment, can develop the sensitivity and specificity needed to identify meaningful signals within the noise of legitimate biological use, including a reduction of both false positives and false negatives to operationally acceptable rates.
- Detection approaches can be developed and deployed in ways that preserve user privacy, and clearly defined and publicly communicated terms govern when and how privacy protections might be overridden for purposes of deploying safeguards or making referrals to law enforcement.
- The effectiveness of misuse monitoring scales with the breadth of data and cross-platform coordination available, making standardization and information-sharing across developers and deployers and with government a condition for maximizing the value of this mitigation.

Mitigation Requirements and Limitations

Effective misuse monitoring requires several distinct investments. Analysis must draw on a sufficiently broad and representative sample of user interaction data—including sessions from legitimate scientists working on virology, immunology, and microbiology—to develop baseline models of normal biological information-seeking behavior against which anomalous patterns can be identified. This requires interdisciplinary collaboration between AI developers, AI deployers (such as cloud service providers), domain-area scientists, and biosecurity specialists who understand both the legitimate landscape of biological inquiry and the specific patterns that might characterize misuse attempts at different stages of the biological risk chain. Such tasks could possibly be automated through the development of a biosecurity AI agent that can evaluate a specific query to assess the level of biosecurity concern.[27] Privacy-preserving tools—including such approaches as differential privacy, federated analysis, and automated flagging systems that do not expose individual user data to human review (absent a defined threshold of concern)—are a prerequisite for this work to proceed in a manner that is acceptable to users and consistent with legal and ethical obligations.

It is worth noting that elements of this mitigation already exist in practice. Some frontier AI companies already conduct monitoring for potential misuse and take enforcement action against users who appear to violate policy. In practice, however, escalation beyond platform enforcement remains limited and often ad hoc. In many cases, suspicious activity might result only in account restriction, service termination, or internal watch listing unless the misuse signal rises to a level that is especially clear, severe, and discernable. This creates an important policy and governance question of whether AI-enabled biological misuse should remain under a largely discretionary reporting model or whether a more formal threshold for referral should be established. One possible model is a narrowly scoped reporting framework that is analogous in structure to existing mandatory-reporting regimes for certain categories of severe online harm, such as child sexual abuse material. Any such framework would require careful line-drawing to (1) define what levels of confidence,

[27] Josh Dettman, Emily Lathrop, Aurelia Attal-Juncqua, Matthew Nicotra, and Allison Berke, "Prioritizing Feasible and Impactful Actions to Enable Secure AI Development and Use in Biology," *Biotechnology and Bioengineering*, Vol. 123, No. 5, May 2026.

capability relevance, and risk severity justify referrals and (2) ensure that stronger reporting expectations do not create unacceptable costs for privacy, civil liberties, legitimate scientific use, or false-positive reporting.

Key features of this mitigation class include (1) privacy-preserving technical architectures for real-time analysis of user interaction data, (2) interdisciplinary frameworks for distinguishing legitimate biological information-seeking from misuse patterns, (3) cross-platform data-sharing and standardization mechanisms supported by coordination infrastructure, (4) clearly defined and publicly communicated terms of use governing privacy and escalation, (5) participation from frontier AI companies and their service providers, and (6) clearly defined escalation and referral pathways, including appropriate law enforcement channels, for high-confidence and credible misuse signals.

This mitigation is effective only against threat actors who require or seek meaningful uplift from closed-weight models. Actors who already possess the requisite expertise and resources, including radicalized insiders or individuals operating within a state's biological weapons program, would not be substantially affected by detection of information-seeking behavior because their primary barrier is not information access. For those actors, the deterrence and physical chokepoint interventions described previously are more relevant. This mitigation is most effective when treated as a component of a broader detection architecture that includes the centralized AI and biology information-sharing nexus that we describe in the following section, through which signals generated by individual developer-monitoring programs can be aggregated, contextualized, and acted on at the ecosystem level.

Centralized AI x Bio Information-Sharing Nexus

Problem Overview

The final mitigation provides a foundation for strengthening each of the previous mitigations across the biological risk chain. The interventions in the strategy that we describe in this report generate their greatest value not at the level of any individual company but in aggregate. A threat actor who probes multiple frontier models, queries multiple synthesis providers, and sources materials across separate vendors might generate signals that are ambiguous or subthreshold at each individual node but would be identifiable as a coherent pattern of concerning behavior if those signals could be combined and analyzed together. A centralized information-sharing nexus—functioning as a clearinghouse for user and customer data across AI developers, synthesis providers, cloud laboratories, and other relevant actors in the biological ecosystem—would close this gap.

How This Mitigation Reduces Risks

By establishing a single entity responsible for aggregating, analyzing, and acting on misuse signals across the AI-biology supply chain, such an architecture would provide situational awareness that no individual actor can achieve alone, creating new opportunities for real-time disruption of nefarious activity and generating the forensic record needed to support attribution investigations after an incident.

For this mitigation to be effective, we assume the following:

- Threat actors seeking to develop AI-enabled biological weapons will interact with multiple nodes of the AI-enabled biological ecosystem across their planning and execution, generating distributed signals that are only actionable when analyzed in combination.
- No individual company or agency currently has the visibility, authority, or data infrastructure to perform this aggregation function, and voluntary bilateral data-sharing arrangements are insufficient substitutes for a purpose-built, centralized entity.

- A centralized nexus—if it is established with appropriate legal authorities, cybersecurity protections, and governance safeguards—can operate in ways that are consistent with privacy rights and civil liberties using privacy-preserving technical architectures.
- The deterrent value of this entity scales with its public visibility: Actors who are aware that their interactions across the ecosystem are subject to aggregated analysis face a meaningfully higher perceived risk of detection.

Mitigation Requirements and Limitations

The technical and institutional requirements for such an entity are substantial and represent the primary implementation challenge for this mitigation class. Such a centralized entity for sharing information across the AI-enabled biology supply chain will require concerted effort and analysis to solve key challenges.

On the technical side, effective aggregation requires the use of standardized data formats across participating organizations so that signals from AI model providers, synthesis companies, and other vendors can be compared and analyzed within a common framework. These standards should govern not only the structure of shared data but also what information is available automatically for analysis versus what requires justification to access. Privacy-preserving technical architectures can make such data-sharing far more appealing to industry participants than traditional centralized models. Participating organizations could run classifiers locally with relatively low thresholds for flagging potential biological misuse, then hash or anonymize personally identifiable information and other sensitive fields before sharing the flagged record with the central entity. The hashed identifiers would still enable cross-platform data fusion—allowing the entity to identify patterns of concerning behavior across providers—without raw personally identifiable information sitting in the centralized dataset by default. Such hashed user identifiers and flagged query patterns might be available by default for pattern matching, while the deanonymization of a flagged user would require a credible threat pattern to emerge and approval under defined protocols. This tiered approach has precedent in other domains in which sensitive data must be shared across organizations for safety purposes, and it substantially reduces the privacy risks inherent in centralized aggregation. Automated-detection pipelines will be essential for the nexus's real-time function, but human expert review will be necessary for cases that require contextual judgment before escalation to law enforcement or other response channels.

On the institutional side, several foundational concerns require deliberate analysis before such an entity can be established. These include the appropriate legal authorities under which it would operate; whether it should be housed within an existing federal agency, established as an independent body, or structured as a public-private partnership; the governance arrangements that would give participating companies confidence that shared data will be protected and used appropriately; the formal connections to law enforcement and intelligence community channels through which actionable findings would be referred; and the public communication strategy that would allow the entity's existence and mandate to serve a deterrent function without revealing operational details that could be exploited to evade detection. International engagement and standardization will also be required because threat actors operating across jurisdictions could exploit gaps between national information-sharing architectures if equivalent entities are not established and coordinated among allied nations.

Key features of this mitigation include (1) near-complete coverage agreements with frontier AI developers and synthesis providers; (2) standardized data formats enabling cross-platform signal integration; (3) robust cybersecurity protections for aggregated data; (4) privacy-preserving technical architectures, including local classification, hashing, tiered access protocols, and industry-specific sharing standards that calibrate data exposure to the sensitivity and operational value of different signal types; (5) expert analytical staff combining software engineering, biological domain knowledge, and intelligence analysis capabilities; (6) formal law enforcement and intelligence community referral channels under appropriate legal and constitutional con-

straints; (7) public communication about the entity's mandate that is sufficient to generate a deterrent effect; and (8) international coordination to prevent jurisdictional gaps from becoming evasion opportunities.

This is perhaps the most institutionally complex and resource-intensive of the interventions that we describe in this report, and it is unlikely to succeed without binding regulatory authorities, dedicated federal resourcing, and sustained engagement with the private sector to build the participation agreements and data-governance frameworks on which the entity depends. More analytical work is needed to define the precise implementation pathway, assess the legal and constitutional constraints that will shape its design, and determine whether and how it should be pursued. The potential returns, including real-time misuse disruption, enhanced attribution capability, and a deterrent effect extending across the full supply chain, make it among the highest-leverage investments in this strategy if those implementation challenges can be resolved.

CHAPTER 3

Comparative Analysis of Strategy Components

The preceding chapter described a layered prevention strategy aimed at reducing the likelihood of pandemic-scale AI-enabled biological incidents through nine distinct mitigations spanning four mechanisms of change. The strategy's value, however, is not located in any single intervention. No individual mitigation is sufficient to interrupt all pathways to misuse across all threat-actor profiles, and each mitigation carries meaningful limitations that other mitigations are designed to partially offset. The strategy's strength is therefore systemic: It works by accumulating friction, cost, and detection probability at multiple points along the biological risk chain rather than by creating an absolute block at any one point. Understanding how this network of mitigations functions as a coherent whole is essential to effective implementation.

This chapter provides that comparative analysis by analyzing how the nine mitigations work together using two primary lenses: coverage across the biological risk pathway—from ideation to weaponization—and coverage across the threat-actor space—from opportunistic lone actors to state-sponsored programs. Within this analysis, we identify key complementarity clusters—groupings of mitigations that are the most mutually reinforcing for key risk-reducing activities—and examine what the strategy loses when any cluster is weakened. This analysis is meant to showcase why and how we chose these mitigations to function together in a coherent strategy.

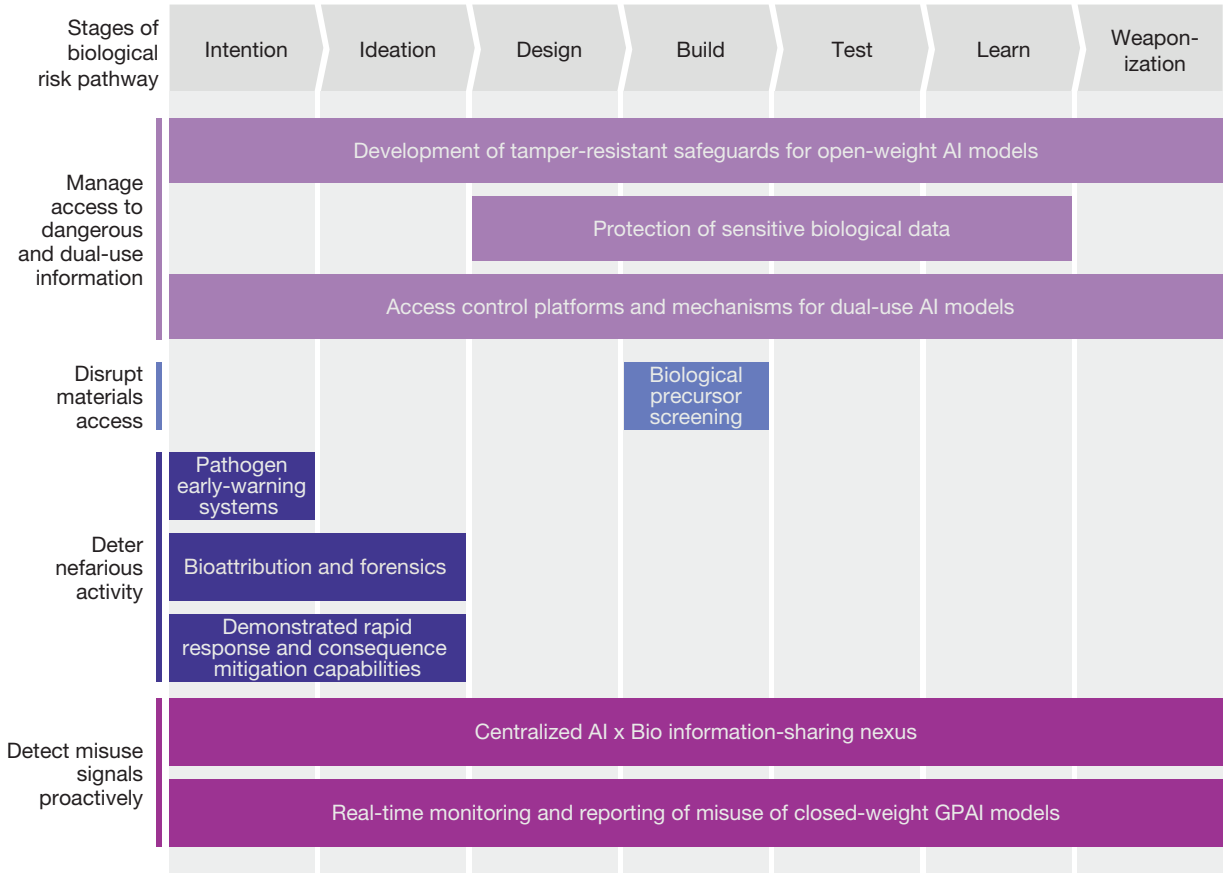
Coverage Across the Risk Pathway

We built the mitigation strategy on the understanding that actors must achieve certain milestones to succeed at developing biological threats. These milestones are (1) developing and sustaining motivation to cause harm with biology, (2) using the appropriate tools and data to ideate and design an enhanced or novel biological threat, (3) accessing the right information to successfully execute the steps to biological threat development and weaponization, (4) obtaining the appropriate precursor materials and equipment to translate designs to physical manifestations, and (5) circumventing biosecurity controls and law enforcement to avoid detection. We built a strategy with interventions that can create friction across the risk pathway's stages with mitigations that are targeted to manage information access, disrupt materials access, deter nefarious activity, and enable the proactive detection of misuse behavior (see Figure 3.1).

This strategy has dense coverage across the risk pathway's stages to align with the layered defense paradigm that is critical for ensuring robust biodefense.[1] The coverage is most dense across the design-test-

[1] Richard Danzig, *Preparing for Catastrophic Bioterrorism: Toward a Long-Term Strategy for Limiting the Risk*, Center for Technology and National Security Policy, May 2008; Bipartisan Commission on Biodefense, *The National Blueprint for Biodefense: Immediate Action Needed to Defend Against Biological Threats*, April 2024; William Beaver, Yong-Bee Lim, Lillian Parr, Christine Parthemore, Andrew Weber, Natasha Bajema, Rohit Chitale, Jackson duPont, Chris Fall, Nikki Teran, and Alexander Titus, *A Handbook for Ending Catastrophic Biological Risks: How the United States Can Prevent Future Pandemics and Deter Biological Weapons*, ed. by Francesco Femia and Christine Parthemore, Janne E. Nolan Center on Strategic Weapons, Council on Strategic Risks, December 2021; Office of the President of the United States, *National Biodefense Strategy*

FIGURE 3.1
Mitigations Mapped Across the Biological Threat Risk Pathway



build-learn cycle primarily because we seek to reduce actors' capacities to effectively design a novel pathogen that could have global consequences, especially regarding the ability to detect or respond to the threat. By concentrating coverage and seeking to bottleneck these design capabilities, we seek to reduce this particular risk pathway through additional friction. Dense coverage across the early stages of the pathway is the goal, and three information-restriction mitigations and precursor screenings provide robust friction at the earliest stages of the risk chain. Disruption or deterrence at these early stages or increasing the perception that biological misuse is challenging leads to reduced risk of misuse—that is, an actor who cannot access the knowledge required to design a novel pathogen or the nucleic acid precursors to construct one will not reach the test, learn, or weaponization stages regardless of their motivation or resources. Early-stage intervention is the strongest leverage point in the chain; therefore, the strategy concentrates its most numerous and overlapping mitigations there.

However, none of these early-stage mitigations will be perfectly implemented: Information controls can be circumvented or are irrelevant for expert actors, synthesis screening has coverage gaps, and access-controlled platforms can be exploited by credentialed insiders or sufficiently resourced actors. In these cases, disruption shifts to additional layers, such as the proactive detection of misuse behavior across the risk chain. Real-time

and Implementation Plan for Countering Biological Threats, Enhancing Pandemic Preparedness, and Achieving Global Health Security, October 2022; Office of the President of the United States, *National Biodefense Strategy*, 2018.

monitoring and a centralized information-sharing nexus are designed to detect actors who have successfully bypassed the early-stage interventions and are exhibiting escalating patterns of misuse behavior.

In finalizing the mitigation set, we also assessed actor-specific strengths and limitations to ensure that no actor profile—regardless of the actor's approach to information access, material acquisition, or operational obfuscation—faces an unobstructed path through the risk chain.

Coverage Across the Threat Actor Space

Using the threat actor profiles established in Chapter 1, we stress tested our strategy by analyzing how the mitigation network performs against each major actor type—a lone actor with limited expertise, radicalized scientists, nonstate groups, and state biological weapons programs—by creating decision trees that map across the various technical, motivational, and operational milestones that an actor must achieve as they move from ideation through the successful weaponization of an AI-enabled pathogen capable of causing mass harm.

We then mapped each mitigation across the milestones to understand where along the risk pathway the plan added meaningful friction to prevent actors from reaching those milestones. We present the overarching decision tree with mitigations mapped across each node in Figure 3.2. The key milestones are getting the initial motivation to do harm with biology, reaching the comfort level for attribution and consequences (e.g., "Wants credit" or "Avoids credit"), selecting the source of the threat design, accessing an AI model to uplift or accelerate efforts, obtaining the pathogen in whole or via nucleic acid, and accessing laboratory resources. Some mitigations are more relevant to certain milestones depending on how they block or provide friction.

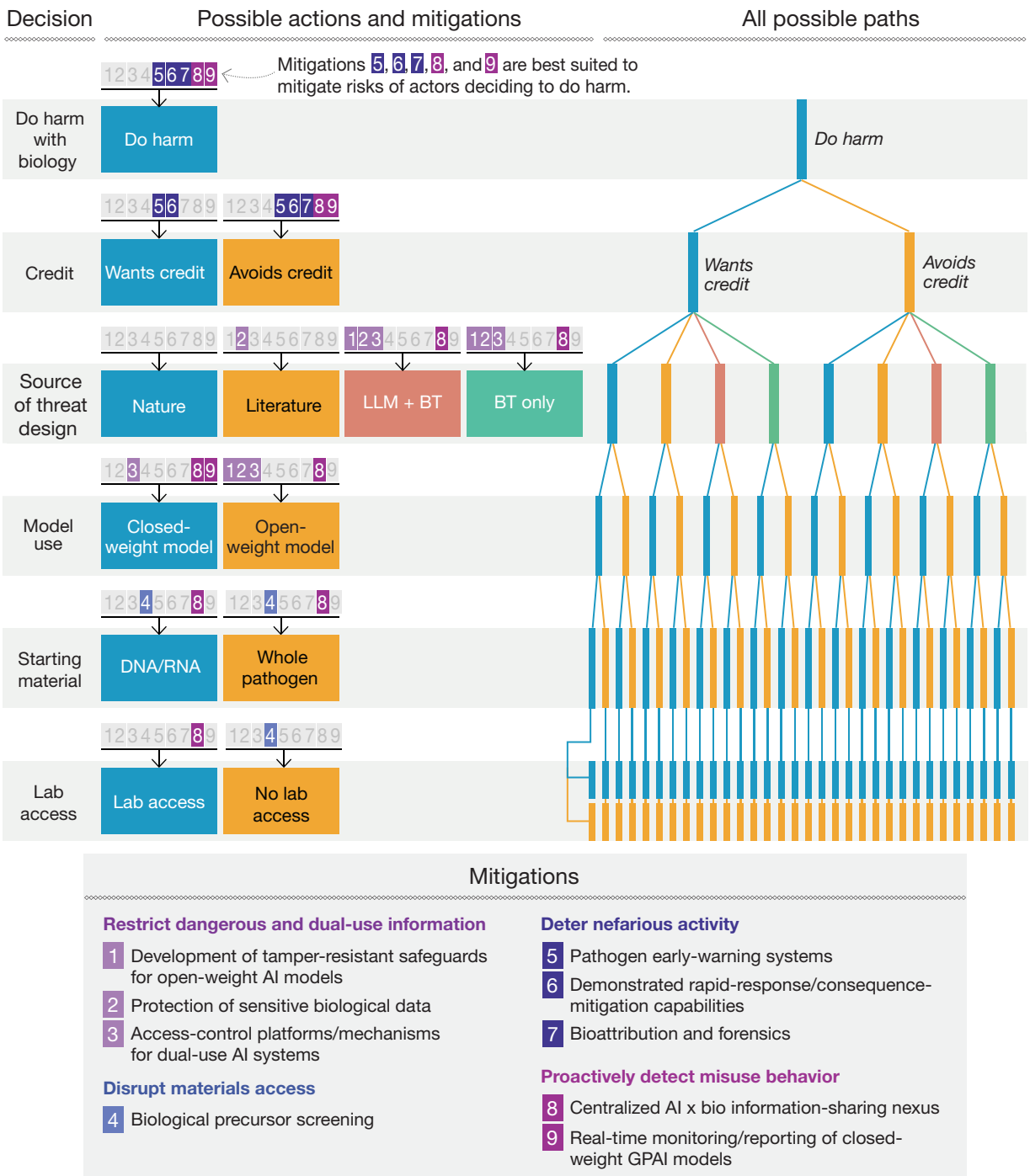
After mapping the network of mitigations across the decision tree, we examined actor-specific pathways through the decision tree using prototype capabilities of each actor class (we note that these provide representative examples to stress test our network of mitigations and might not be comprehensive). Two paths to harm are shown in Figure 3.3. We held iterative discussions to decide actor decisions and where our mitigation strategies are effective along the pathway. The pathway is meant to showcase representative decisions that certain actor profiles might make and to stress test how our mitigations provide coverage across diverse paths to harm. A central finding from this analysis is that different actor profiles are susceptible to different mitigation classes. Some actors appear relatively amenable to disruption but difficult to deter, while others appear difficult to disrupt but more susceptible to deterrence. This trade-off motivates a layered strategy that combines both approaches rather than relying on any single intervention.

Disruption-oriented mitigations are generally most effective against actors who lack expertise, resources, access, or institutional legitimacy. Some interventions, including open-weight safeguards, access-control platforms, and real-time monitoring, are particularly effective against actors who rely on AI systems to compensate for their limited biological expertise. Other mitigations, including precursor screening, biological data protection, and access controls on sensitive data, are designed to constrain actors who lack access to the materials, equipment, and institutional resources needed to execute biological research. Although these categories overlap substantially today, they address different capability dimensions and might become increasingly complementary if advances in AI reduce expertise barriers while resource constraints remain binding. Collectively, these measures are likely to be most effective against lower-resourced individuals and groups and less effective against actors who already possess privileged access, advanced expertise, or extensive infrastructure.

The opposite side of this trade-off concerns actors who are difficult to disrupt directly. For example, highly capable state-sponsored programs and, in some cases, well-resourced nonstate actors might possess the expertise, infrastructure, and access needed to bypass many information and material access controls. For these actors, deterrence becomes increasingly important. Attribution capabilities, early-warning sys-

FIGURE 3.2

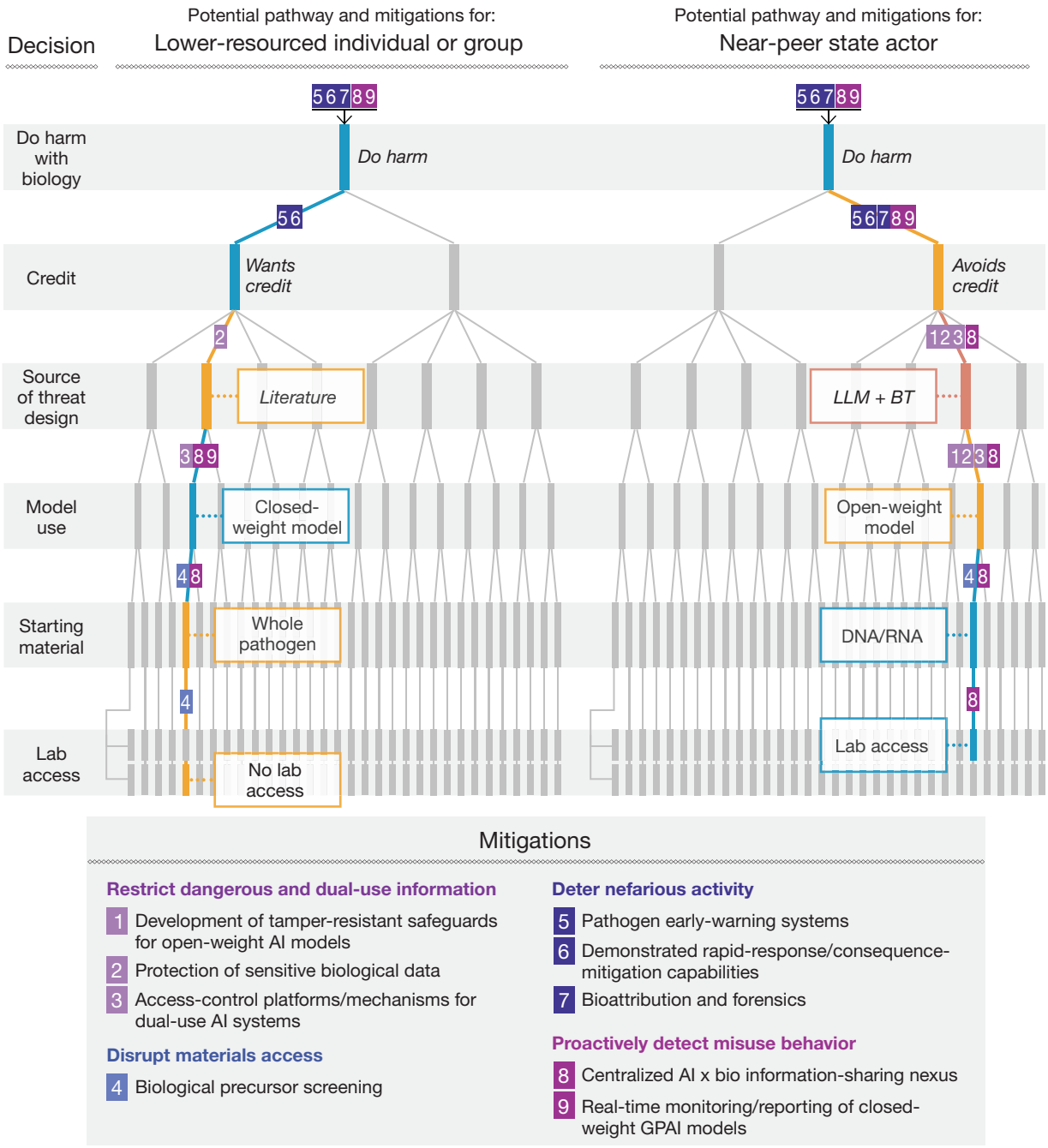
Mitigations Mapped Across Milestones in an Actor Decision Tree



tems, and demonstrated rapid-response and consequence-mitigation capabilities can alter the expected costs and benefits of an attack even when direct disruption is unlikely. Conversely, some ideologically motivated actors might be less responsive to deterrent signals, increasing the importance of disruption-oriented measures, monitoring, and detection for those cases.

FIGURE 3.3

Representative Paths to Harm for Diverse Threat Actors



These findings suggest that a strategy that pairs disruption-oriented mitigations with credible deterrence-oriented backstops is more robust than one that concentrates investment at any single point of the risk pathway. Adding a layer of mitigations that is meant to enable the proactive detection of misuse can be helpful for both disruption and deterrence mechanisms across the threat-actor space. This network of mitigations forces any actor who successfully navigates one layer to immediately confront the next, and the network ultimately succeeds by accumulating friction, cost, and detection probabilities at multiple points rather than blocking any single pathway absolutely.

Key Complementarity Clusters

Although all nine mitigations contribute independently to the strategy's prevention objectives, several mitigations are tightly grouped to achieve meaningful risk-reduction activities. Their joint implementation produces substantially greater protective value than any mitigation would achieve alone, and weakening one mitigation within a cluster degrades the effectiveness of the others. Three such complementarity clusters emerged from our analysis, as follows:

- **Cluster 1—Reduce the capacity to execute AI-enabled design workflows for novel biological weapon design** (open-weight safeguards + biological data protection + access-control platforms). This set of mitigations is specifically targeted to reduce the ability of actors of all types in their quest to design, develop, and deploy a novel biological threat. By reducing access to the data and tools that have the highest potential for misuse to only legitimate users, achieving breakthroughs with limited risks is possible. Biological data protection provides a useful foundation in this cluster because if sensitive data are widely accessible, they can be used to enhance open-weight models' dangerous capabilities, and/or LLM agents could use them to design BTs from scratch that have capabilities similar to those behind access controls, thereby obfuscating the utility of open-weight safeguards and access-control platforms.
- **Cluster 2—Enhance proactive misuse detection through reinforced information and material access controls** (precursor screening + real-time monitoring + centralized information-sharing nexus). The foundation of this cluster is the centralized information-sharing nexus, which enhances the likelihood of achieving deterrence and disruption from monitoring and screening mechanisms that precursor providers and information provisioners (e.g., AI developers) implement by providing a mechanism to integrate signals and connect with government authorities.
- **Cluster 3—Create a compounding deterrence signal through a credible detection and consequence architecture** (bioattribution + early warning + rapid-response infrastructure). Some actor profiles might be difficult to disrupt through informational and material access control alone. For these actors, deterrence becomes increasingly important, and this is where attribution capabilities, early-warning systems, and rapid-response infrastructure can work together to increase the expected cost of an attack while reducing the confidence that desired effects can be achieved. For this, a credible attribution capability is necessary, which requires both technological advancement and a concerted policy effort. Credible attribution strengthens deterrence by consequence by increasing the likelihood that actors will believe that they will be identified and face consequences for misuse. Such attribution capabilities are strengthened by robust early-warning systems because earlier detection generates more information and evidence that can support high-confidence attribution. Rapid-response and consequence-mitigation capabilities further strengthen the deterrence signal by reducing confidence that a biological attack will achieve meaningful impact, thereby contributing to deterrence by denial. Together, these capabilities create a more comprehensive deterrence architecture that addresses both the expected costs and the expected benefits of an attack.

These illustrative clusters define the key efforts we hope that our strategy will achieve. The cluster analysis also revealed two mitigations within this strategy that we consider foundational, meaning that their absence would reduce the effectiveness of several other mitigations. The first is the use of a centralized information-sharing nexus, which can integrate signals derived from most of the other eight mitigations and, if resourced appropriately, could serve as a strong disruption and deterrent feature of the biosecurity enterprise. The second is precursor screening, which is the only mitigation that increases effort at the digital-to-physical divide. An emerging foundational mitigation might be biological data protection because such protection could reinforce and prevent the enhancement of both GPAI and BT capabilities to dangerous capability levels.

However, more work is needed to better understand how robust this intervention can be as AI-enabled biological capabilities rapidly accelerate.[2]

Understanding these clusters is essential for knowing how to prioritize the strategy's implementation. Such an understanding reveals how gaps in a single mitigation can create systemic vulnerabilities that extend well beyond that mitigation's individual contribution. We discuss further implementation considerations in the next chapter, and we give a particular focus on each mitigation's readiness.

[2] Three key questions have emerged from our analysis that require more study: whether (1) laboratory-validated experimental data will remain dramatically more useful than synthetic biological data that emerges from virtual experimentation, (2) more-powerful BTs continue to require fine-tuning to adequately perform specific tasks or whether their capabilities will generalize even across potentially dangerous capabilities of concern, and (3) the amount and quality of currently available public data are insufficient to train generalizable models with capabilities of concern. If any of these status quos shift or flip, the protection of biological data becomes less useful.

CHAPTER 4

Mitigation Readiness

Although in-depth analyses and the development of implementation plans are outside this report's scope, in this chapter, we provide a preliminary look at future implementation by assessing each mitigation's readiness and examining the cross-cutting implementation challenges each will face. We organize the chapter around three variables central to the strategy's real-world viability: the regulatory environment, the state of international cooperation, and AI progress dynamics. We then provide preliminary guidance on how to sequence the implementation for short- and medium-term implementation.

This strategy's nine mitigations depend on both the technical maturity and the implementation realities of various regulatory, international, and technological dimensions. The mitigations fall into four primary technical feasibility levels: nascent with more research needed, technically feasible but early development, established approach but implementation gaps remain, and technically tractable but institutionally complex. Table 4.1 categorizes the different mitigations within these levels, highlighting how the overall strategy consists of a balanced mix of early-stage research bets and established approaches.

Notably, three of the mitigations are considered technically tractable but institutionally complex, shifting the implementation burden of success not onto technical needs but rather onto other factors related to proactive leadership and resource prioritization. On the other side of the spectrum, the most nascent of the mitigations are also some of the potentially most impactful: The successful development of tamper-resistant safeguards for open-weight models would reduce a currently unmanaged misuse risk, and robust bioattribution—especially techniques that are sequence-based—could serve as a powerful deterrent against a class of actors that is challenging to disrupt. Quickly understanding whether each mitigation is possible to achieve will have implications for the rest of the strategy's robustness of and decisions for complementary post-release resilience mitigations. The strategy's value is not in any single mitigation but in the cumulative friction, cost, and detection probabilities it creates across different iterations of the risk chain. However, the success or failure of one node will affect the overall robustness. Besides these considerations related to technical feasibility, each of these mitigations will rely on the context through which they are implemented, including the regulatory environment, international cooperation, and AI progress.

Regulatory Environment

The control and management of AI risks from GPAI systems and BTs have thus far largely been industry- and academic-led with limited government regulation.[1] In this environment, the U.S. government has taken the role of standard setter and industry convener, taking its cues from the needs of those who are developing this transformational technology rather than investing significantly in regulation, resources, or incentives to

[1] For GPAI systems, the vast majority of risk control and risk management falls to the leading frontier AI companies and their deployment partners. For BTs, the ecosystem is more distributed: Leading capabilities in biodesign and laboratory tasks are derived from tools that industry and academic developers created. This makes risk management even more challenging.

TABLE 4.1
Technical Feasibility of Each Strategy Mitigation

Technical Feasibility Level[a]	Mitigation
Nascent with more research needed	• Development of tamper-resistant safeguards for open-weight AI models • Bioattribution and forensics
Technically feasible but early development	• Pathogen early-warning systems • Real-time monitoring and reporting of misuse of closed-weight GPAI models • Demonstrated rapid-response and consequence-mitigation capabilities
Established approach but implementation gaps remain	• Biological precursor screening
Technically tractable but institutionally complex	• Access-control platforms and mechanisms for dual-use AI systems • Protection of sensitive biological data • Centralized AI x Bio information-sharing nexus

[a] The technical feasibility level was determined using team members' expertise and through iterative team discussions.

encourage or compel industry action. Many of the mitigations we suggest can be implemented by industry or academia without government involvement, but most rely on government resources or policy for maximum robustness. Understanding which mitigation requires which type of government involvement will be important as implementation roadmaps are built for each of these interventions.

The most necessary example of government reliance lies with the centralized information-sharing nexus. Such an institution cannot be developed without government touchpoints, resourcing, and policy that allow for privacy-preserving information-sharing, compel industry to report, and connect meaningful signals of misuse to law enforcement. Without proactive government leadership, such an intervention would not be possible.

Other mitigations—such as real-time monitoring of GPAI, protecting sensitive biological data, or biological precursor screening—could be implemented robustly by the private sector without government regulation, but uniform and consistent implementation might rely on government policy and resources. The demonstrated development and deployment of rapid-response and consequence-mitigation capabilities will likely require active partnerships between government and industry stakeholders to ensure sustainable investments. This mitigation is likely best categorized as technically feasible but early development because the underlying platform science is real but the demonstrated, end-to-end, pathogen-agnostic response capability needed for deterrence remains incomplete across the relevant threat space. Lastly, early-warning systems, such as wastewater surveillance, are being developed throughout the private sector but will ultimately require government resources for maintenance and/or government touchpoints for reporting.

International Cooperation

Many of this strategy's mitigations are less robust if the United States implements them without the cooperation of international allies. For example, pathogen early-warning systems that are only targeted within U.S. borders will very likely miss an opportunity for the earliest warning. Centralized information-sharing will be most effective should international entities also participate to identify early patterns of misuse. Furthermore, both the United States and China are global leaders in AI development and biotechnology progress.[2] Therefore, any mitigation that seeks to reduce information or materials access in these fields will remain

[2] Toby Webster, Richard Moulange, Barbara Del Castello, James Walker, Sana Zakaria, and Cassidy Nelson, *Global Risk Index for AI-Enabled Biological Tools*, Centre for Long-Term Resilience, September 9, 2025.

significantly weakened should the United States implement them alone. The development and adoption of biological data protection, tamper-resistant safeguards for open-weight models, access-control platforms, precursor screening, and real-time monitoring of GPAI models will be most robust if both the United States and China implement them consistently and cooperatively. An enabling condition for this strategy will be dialogue, exchange, and cooperation with China.

Trajectory of AI Progress

The trajectory of AI progress adds urgency to this plan's implementation. Experts in the AI ecosystem have expressed great uncertainty as to when and how AI capabilities will reach superintelligence levels in a particular domain or across multiple domains. Experts predict such a development of artificial general intelligence on the order of months, years, or decades, and some argue it will not develop at all. Although the biological misuse risk might not correlate exactly with the arrival of artificial general intelligence or artificial superintelligence, AI capabilities in the biological domain have been increasing dramatically, and preparing now for the prevention of misuse is prudent.[3]

Preliminary Roadmap to Strategy Execution

Different stakeholders are responsible for the mitigations that we described in this strategy. The early-stage efforts require a dedicated research focus, the development and implementation of mid-stage interventions require government and industry cooperation, and the implementation of the technically tractable mitigations require all three: academia, government, and industry. It will be challenging but necessary to build all mitigations simultaneously. Although robust implementation plans fall outside the scope of this report, we provide some initial prioritization on next steps using the potential impact of the mitigations.

In the immediate term (the next one to two years), nascent technical efforts should be funded to determine which will likely succeed or fail in managing unmanaged risk. Efforts include developing both tamper-resistant safeguards and attribution technology. Simultaneously, actions that are technically ready, are scalable, and do not require new regulatory architecture—including strengthening biological precursor screening, developing access-control minimum standards and platforms, implementing real-time monitoring of GPAI misuse, and expanding pathogen early-warning system coverage—should continue to progress with a goal of scaling up implementation by early 2028. Given the potential for the protection of sensitive biological data to provide a barrier against the design of novel and enhanced biological weapons, efforts to better understand and develop incentives and/or regulations for this intervention should be heavily prioritized.

In the near term (the next two to four years), actions that require foundational research, institutional design, or regulatory development should be a focus—dedicated attention should be on removing barriers to implementation success. This includes the deployment of tamper-resistant open-weight safeguards (should research prove this technique is feasible and robust), the protection of sensitive biological data, and development of the centralized information-sharing nexus.

On a farther-term horizon (greater than four years), actions that depend on prior investments and international coordination should take full operationalization, including the centralized information-sharing nexus and international data-governance regulatory frameworks.

[3] Pannu, Bloomfield, et al., 2025.

Significant efforts will be required to understand the dependencies and deployment steps for the complex mitigations that we have proposed; doing so will aid in developing an implementation framework for not only these discrete mitigations but also their integration into a comprehensive strategy to prevent AI-enabled biological incidents. Such a framework should include timelines and triggers for when implementing certain mitigations is recommended versus critical technical requirements. The strategy and the implementation details can be strengthened by insights from prototyping technical mitigations and red-teaming exercises that assess the gaps and strengths of this combination of mitigations in different threat scenarios. Rigorous evaluations should continue alongside the deployment of new models, AI-biotechnology integration, and mitigation implementation, including tracking the assumptions made throughout this report.

CHAPTER 5

Enabling Conditions and a Path Forward

This strategy focuses on the prevention of high-consequence biological incidents perpetrated by actors using AI-enabled uplift, acceleration, and/or novelty. Any actor seeking to execute an AI-enabled, pandemic-level biological attack must overcome a sequence of barriers across the biological risk chain. These include (1) developing and sustaining the motivation to cause harm with biology, (2) using the appropriate tools and data to ideate and design an enhanced or novel biological threat, (3) accessing the right information and expertise to successfully develop and weaponize the biological threat, (4) obtaining the appropriate precursor materials and equipment to translate designs to physical manifestations, and (5) circumventing biosecurity controls and law enforcement to avoid detection.

This strategy works by reinforcing those barriers at multiple points simultaneously, accumulating the friction, cost, and detection probability required to extend timelines, deter nefarious intentions, and raise the likelihood that suspicious behavior is disrupted before a threat's release. We described combinations of mitigations across a risk pathway, and the mitigations are robust enough to account for the strengths and weaknesses of various threat actors with shifting capabilities and intents. AI capabilities are advancing in ways that alter the feasibility of global-scale biological incidents, and the window for building preventive infrastructure is time-sensitive.

The mitigations proposed for this strategy are ambitious, and implementation will require collaboration between both government and private-sector stakeholders. We conclude this report by describing the key enabling conditions that will be required for the successful execution of this strategy, as well as the remaining gaps not yet addressed.

Enabling Conditions

Key enabling conditions for this strategy include the following:

- **There is credible and sustained U.S. government leadership.** Across nearly every mitigation, federal government action—whether regulatory, investment, signaling, or coordination—is a necessary condition for effectiveness at scale.
- **Institutional governance addresses insider risk, radicalization dynamics, and the trust-accountability balance.** The nine mitigations address technical barriers but not the institutional conditions that allow insider exploitation or radicalization to proceed undetected within legitimate systems. Monitoring for these behavioral signals raises genuine privacy, civil liberty, and legal concerns, and overly intrusive approaches risk eroding the public trust (on which this strategy's legitimacy depends). An effective strategy must therefore balance targeted oversight with the preservation of institutional legitimacy and may leverage and expand existing biosafety programs, such as bystander training. Mitigations must therefore account for both individual-level and structural dynamics.

- **Sufficient international coordination closes jurisdictional gaps.** Several mitigations are materially weakened by unilateral U.S. action. Cooperative action with China and other world leaders in AI development and biotechnology will be necessary to ensure robust risk reductions.
- **A resilience backstop complements prevention activities.** Although this strategy emphasizes prevention before a biological threat is released, preparedness for a post-release response remains an essential component of overall bioresilience. No prevention system can eliminate risk entirely, and high-consequence biological incidents require the capacity to rapidly detect, characterize, and respond to emerging threats. PPE, MCM development, and robust public health systems together serve as a resilient backstop that can reduce harm and guide public health systems' security responses if an incident occurs.
- **Built-in adaptation mechanisms address adversarial dynamics.** This strategy is not static because priority threat actors, AI capabilities, geopolitical dynamics, and biological research methods will all evolve, creating ongoing pressure on each mitigation's effectiveness. The key to success will be an adaptable set of mitigations that includes regular assessments via institutional red-teaming, regular review cycles, and investments in adaptive screening methods so that the strategy remains robust. Mitigations that are most exposed to an adversarial attack are those in which a digital attack or capability drives tampering, and circumvention includes synthesis screening,[1] access controls and biological data protection, tamper-resistant safeguards, and real-time monitoring. Importantly, a recognition of the adversarial dynamic's asymmetry must remain front of mind: Defenders seek to maintain the entire network of mitigations, but malicious actors need to find only one weakness.[2]

Path Forward

The main indicator for this strategy's success will be that no AI-enabled biological attacks are attempted or executed in the foreseeable future, especially as AI continues to progress and enable improvements and benefits in the biological sciences. Obviously, indicators of such an outcome are difficult to measure robustly. Aside from technical and policy measures that demonstrate the successful deployment and integration of these nine prioritized mitigations, results from red-teaming exercises and rigorous evaluations might be necessary to demonstrate whether this strategy is indeed reducing the risk of AI-enabled biological attacks. Intelligence or indicators that these mitigations disrupted a threat actor's attempt at a biological attack, from ideation all the way through weaponization, would also be an indicator of success. Success will require comprehensive and complementary solutions and significant collaboration from both government and private-sector stakeholders to keep up with or exceed the pace of technical developments that affect AI-enabled biological materials and technology and threat actors' calculi and threat cycles. Relaying the importance of these mitigations to members of the scientific community is also important because their legitimate use of AI for benevolent scientific advancements could be affected by additional model safeguards.

Success will also require a careful assessment of the implementation trade-offs, including the potential costs to beneficial research and development, so that the safeguards can be designed in ways that meaningfully reduce risk without unnecessarily constraining legitimate scientific work. There is an urgent and unique window of opportunity for proactive action while advanced AI-enabled biological capabilities are not available or accessible for widespread misuse.

[1] Wittmann et al., 2025.

[2] Aveggio et al., 2025.

APPENDIX A

Summary Table Describing the Strategy's Nine Mitigations

We present a summary of our strategy's nine mitigations in Table A.1. We describe each mitigation and list the mitigation class, risk pathway stage, and threat actor profile associated with the mitigation. We also give synopses of the key assumptions we made when developing the mitigations and each mitigation's limitations.

TABLE A.1

Mitigations Summary

Mitigation	Brief Description	Mitigation Class	Risk Pathway Stage(s)	Key Assumptions	Threat Actors Most Affected	Key Limitations
Development of tamper-resistant safeguards for open-weight AI models	Invest in technical approaches to erode hazardous base capabilities during or before model training, making open-weight models resistant to adversarial fine-tuning while preserving their beneficial utility	Manage access to dangerous and dual-use information	Intention, ideation, design, build, test, learn, weaponization	Open-weight models will eventually approach frontier closed-weight uplift risks; complete release restriction is unlikely without binding regulation; safeguards must preserve legitimate utility to achieve sufficient community adoption.	Moderately capable nonstate actors; technically skilled lone actors; actors seeking to fine-tune models on dangerous data	Scalability and robustness of pretraining approaches remain unvalidated at the frontier scale; adoption is voluntary absent regulation
Protection of sensitive biological data	Restrict access to laboratory-validated hazardous biological datasets through managed-access platforms, publication norms, and secure repositories to deny adversaries data needed to build more-capable AI models	Manage access to dangerous and dual-use information	Design, build, test, learn	Certain specialized lab-validated datasets could meaningfully uplift AI-enabled bioweapon design beyond what public data allow; a time-sensitive window exists before in silico methods could substitute for lab-validated data.	Moderately capable actors seeking to fine-tune AI models; individual actors without independent lab capacity	Requires broad consensus on what data qualify as high risk; tension with open science norms; less effective against state programs capable of generating data independently; become obsolete if in silico biological data become just as useful for fine-tuning as lab-validated data
Access-control platforms and mechanisms for dual-use AI systems	Implement tiered credentialing, verified purpose requirements, time-limited access, and audit and monitoring infrastructure for GPAI models and BTs	Manage access to dangerous and dual-use information	Intention, ideation, design, build, test, learn, weaponization	Expert-level biological AI capabilities meaningfully lower barriers for actors lacking specialized expertise; beneficial use is valuable enough that total restriction is unacceptable; tiered architecture can reduce misuse risk without foreclosing legitimate research.	Moderately capable nonstate groups; lone actors and others without preexisting biological expertise	Does not address actors for whom information access is not a primary barrier; insider threats require additional provisions beyond initial credentialing; coverage is uneven without binding and universal standards

Table A.1—Continued

Mitigation	Brief Description	Mitigation Class	Risk Pathway Stage(s)	Key Assumptions	Threat Actors Most Affected	Key Limitations
Biological precursor screening	Strengthen and universalize synthesis screening across all relevant commercial providers (e.g., gene synthesis, repositories, contract research organizations, cloud labs) using shared threat databases, know-your-customer standards, and direct law enforcement coordination	Disrupt materials access	Build	High-consequence incidents require physical precursors that exceed most actors' in-house synthesis capabilities; commercial synthesis channels are a major chokepoint for nonstate actors; functional inference screening can address AI-designed sequences engineered to evade homology-based detection.	Lone actors; nonstate groups with limited resources; moderately capable actors relying on commercial precursor channels	Ineffective against state programs or others with indigenous synthesis capabilities; ongoing adversarial dynamic between screening methods and AI-assisted sequence obfuscation; coverage gaps persist until all provider types are included; benchtop synthesizer proliferation creates a challenge
Bioattribution and forensics	Develop integrated biological forensic and digital forensic capabilities to credibly link attacks or AI-enabled misuse to specific actors, paired with the U.S. government's consequence doctrine and international coordination to translate attribution into meaningful costs	Deter nefarious activity	Intention, ideation	Some actors are deterred by the prospect of attribution and downstream consequences; attribution capability functions as a deterrent, even if imperfect, when believed to be mature and exercised; effective attribution requires the integration of biological and digital forensic tracks.	State-sponsored programs; state-adjacent and proxy actors; ideologically motivated nonstate groups with reputational or political stakes	Technically nascent with limited validated methods for AI-assisted pathogen attribution; sophisticated state actors might operate in ways that evade attribution; deterrent value depends on the U.S. government's credibility and willingness to exercise consequences following attribution; some actors remain undeterred regardless of exposure risk
Pathogen early-warning systems	Deploy a passive, continuously operating, pathogen-agnostic environmental surveillance network to detect biological threats before clinical recognition, thereby degrading attacker confidence in avoiding attribution and consequences	Deter nefarious activity	Intention	Some actors are deterred when the probability of rapid detection and attribution is perceived to be high; a pathogen early-warning system provides earlier warning than clinical or syndromic surveillance; an early warning generates a forensic signal that supports downstream attribution.	State and nonstate groups with deterrable motivations	Significant infrastructure investment is required for adequate coverage; targeted or localized release might fall below detection thresholds; deterrent value requires public credibility of the systems' maturity

Table A.1—Continued

Mitigation	Brief Description	Mitigation Class	Risk Pathway Stage(s)	Key Assumptions	Threat Actors Most Affected	Key Limitations
Demonstrated rapid-response and consequence-mitigation capabilities	Develop and repeatedly demonstrate the effectiveness of responsible capabilities that can rapidly target a broad variety of biological threats, thereby reducing actors' confidence that an attack will achieve their desired scope of harm	Deter nefarious activity	Intention, ideation	Some actors are deterred when they believe that an effective response will limit their desired scope of harm; platform-based and/or pathogen-agnostic countermeasures and protective measures serve as a meaningful deterrent when they can be quickly adapted across a variety of biological threats; effective deterrence depends on validated real-world performance rather than only theoretical potential.	Broad variety of actor types with deterrable motivations	Some biological threats might still spread faster than countermeasures can be developed and deployed; some biological threats might be incompatible or resistant to existing platform technology approaches; real-world constraints, including manufacturing capacity limits, distribution challenges, and insufficient public uptake might limit effectiveness
Real-time monitoring and reporting of misuse of closed-weight GPAI models	Develop privacy-preserving technical architectures for real-time analysis of user interaction patterns across closed-weight model sessions to detect early misuse signals and generate timely law enforcement referrals before actors advance along the biological risk chain	Detect misuse signals proactively	Intention, ideation, design, build, test, learn, weaponization	Safeguards of frontier closed-weight models will remain susceptible to circumvention despite continued investments in safeguarding, making detection a necessary complement to prevention; real-time misuse detection is technically tractable with focused investment; privacy-preserving architectures are feasible and legally appropriate; effectiveness scales with cross-platform coordination.	Actors relying on closed-weight model uplift for ideation, development, and/or planning; moderately capable nonstate actors; opportunistic actors without preexisting biological expertise	Does not affect actors with sufficient preexisting expertise who do not require frontier closed-weight model uplift; false-positive rates risk restricting legitimate researchers; requires industry standardization and data-sharing to realize full value; most effective only in combination with the centralized information-sharing nexus

Table A.1—Continued

Mitigation	Brief Description	Mitigation Class	Risk Pathway Stage(s)	Key Assumptions	Threat Actors Most Affected	Key Limitations
Centralized AI x Bio information-sharing nexus	Establish a centralized clearinghouse aggregating misuse signals across AI developers, synthesis providers, cloud labs, and other AI x Bio ecosystem actors to enable cross-platform pattern detection, real-time disruption, enhanced attribution, and a deterrent effect through known coverage	Detect misuse signals proactively	Cross-cutting; aggregates signals across the full prerelease risk chain	Threat actors interacting with multiple ecosystem nodes generate distributed signals that are only actionable in aggregate; no single entity has sufficient visibility or authority to perform this function; a centralized nexus can operate within privacy and civil liberty constraints with appropriate governance; the deterrent value scales with public visibility of the entity's mandate.	Broad variety of actor types who interact with multiple components of the AI and biology ecosystem across planning and execution	Most institutionally complex and resource-intensive mitigation in the strategy; requires binding regulatory authorities and dedicated federal resourcing; substantial unresolved legal, constitutional, and governance questions; requires international coordination to prevent jurisdictional evasion

APPENDIX B

AI-Enabled Biological Threat Risk Pathway

In this appendix, we provide a detailed breakdown of each stage in the AI-enabled biological threat risk pathway. We include definitions, illustrative examples, and an analysis of how AI systems might increase an actor's capability (*uplift*) at each step. We also discuss the iterative nature of the life cycle and its grounding in prior literature on terrorism planning and biological weapons development.

AI-Enabled Biological Threat Risk Pathway

The AI-enabled biological threat risk pathway we considered for this strategy and assessed our mitigations against identifies the key steps that a threat actor must complete before releasing a biological weapon. Although this is a generalized risk pathway, and different threat actors might approach these stages differently or add or skip a step, this risk pathway outlines the main steps needed to develop a biological weapon. AI might differentially affect an actor's success at each of these steps to shape the current and future risk landscapes.[1]

This risk pathway integrates critical stages from literature on terrorist planning, biological weapons development, and synthetic biology. U.S. counterterrorism agencies outline phases that are common across terrorist-planning life cycles, such as target selection, surveillance, planning, rehearsal, execution, and escape.[2] Literature on state biological weapon programs and nonstate bioterrorism capabilities introduce other phases that are relevant to developing and handling biological weapons. For example, one framework describes bioweapon development through the five stages of research, scale-up, production, testing, and weaponization.[3] The preattack stages of a terrorist's ability to employ biological weapons have similarly been represented as an engineering process, including acquiring, processing, weaponizing, planning, and deploying.[4]

Prior assessments of how AI models affect the risk of biological weapons that are rooted in historical state- and nonstate efforts emphasize a linear model in which each step represents a potential stopping point on an obstacle.[5] However, a more realistic depiction includes repeated iteration, especially for the stages beyond ideation (described below). In other words, weaponization is simplified if threat actors can try again, especially with help from AI.[6] Therefore, the AI-enabled biothreat risk pathway we developed for this strategy integrates this feedback loop. The design-build-test-learn cycle is an iterative process used in engineering

[1] Rose and Nelson, 2023.

[2] Office of the Director of National Intelligence, undated; U.S. Army, undated.

[3] Sonia Ben Ouagrham-Gormley, *Barriers to Bioweapons: The Challenges of Expertise and Organization for Weapons Development*, Cornell University Press, 2014, p. 41.

[4] Gerstein, 2009, p. 156.

[5] Brent and McKelvey, 2025.

[6] Brent and McKelvey, 2025.

processes to develop and improve systems, and the ideation of a particular system typically precedes the first design step.[7] The scope of this report integrates these insights along with a shift away from an acquisition-focused risk assessment of a biological threat in a technical environment in which AI-enabled design and development are of key concern.

Therefore, the risk pathway we considered for this strategy has seven steps, each of which is critical for AI-enabled biological threats. This risk pathway was outlined by the Centre for Long-Term Resilience,[8] and RAND researchers have adapted it to develop a risk assessment.[9] AI affects all these steps, and GPAI and AI-enabled BTs present varying levels of operational and biological risk. We did not include the "release" step in our risk pathway because of our focus on left-of-boom requirements.

We considered the uplift that AI-enabled BTs and GPAI systems enable for each of these steps. *Uplift* refers to an increase in skills that AI use provides to human users of varying expertise and skill levels. As noted in previous studies and AI safety evaluations, this increase in skills might allow a user to complete a task more quickly and more effectively or might raise the ceiling of dangerous outcomes from AI use.

Stage 1: Intention

We define *intention* as an actor's initial motivation to pursue a biological solution to a strategic aim or grievance. Biological weapons have historically been rarely used. However, AI changes this calculus for threat actors by lowering perceived thresholds to actors who previously viewed access to bioweapons as out of reach. Even if the required tacit and technical capabilities exceed a given actor's capabilities, the perceived ease of obtaining AI-enabled bioweapons might lower the bar to intention and subsequent steps in the biological risk chain for a greater number of actors.

Stage 2: Ideation

We define *ideation* as the determination of a suitable biological attack plan with a particular emphasis on pathogen selection, converting abstract intent into a specific technical concept that supports a goal. In the same way that the ideation step focuses the vague intention into a technical concept, GPAI models synthesize widespread biological information to propose viable attack vectors. Although closed-weight LLMs restrict harmful queries and returns about these topics using various methods, the dual-use character of biological knowledge makes perfect sanitization impossible, and these guardrails are not easily enforced for increasingly open-source GPAI systems. Although previous-generation GPAI systems present only a mild uplift for biological threat ideation, these barriers might be decreasing rapidly, and GPAIs generally make existing capabilities accessible to a larger number of individuals.[10]

Stage 3: Design

We define *design* as the use of synthetic biology and AI tools to create novel sequences and structures. AI-enabled BTs could increase the ceiling of capabilities for both skilled and unskilled actors, particularly when

[7] National Academies of Sciences, Engineering, and Medicine, 2025b.

[8] Rose and Nelson, 2023.

[9] Brady et al., 2026.

[10] AI Security Institute, *International AI Safety Report 2026*, February 3, 2026; Götting et al., 2025; Bill Drexel and Caleb Withers, *AI and the Evolution of Biological National Security Risks*, Center for a New American Security, August 2024.

AI agents can be leveraged to utilize BTs for an actor.[11] For example, dual-use specialized tools can optimize pathogens for such traits as immune evasion or stability.[12] AI integration is particularly concerning for this step because design tools could be used to develop novel structures that evade biosecurity screening capabilities by creating functional homologs with different sequences.[13]

Stage 4: Build

We define *build* as the synthesis of biological material, or crossing the "digital-physical divide." GPAI systems might be able to affect risk at this phase by providing laboratory protocols and other increasingly multimodal instructions to assist actors in physically creating harmful biological material or refining nonpathogenic material to meet their aims. LLMs likely provide the most significant delta for actors with limited expertise to begin with, but more work is needed to better define the nature of any uplift that AI agents can provide to expert actors. This build phase has historically been the stage at which biological weapons programs fail and represents a key entry point for material controls.

Stage 5: Test

We define *test* as the verification that the threat actor's pathogen functions as intended. GPAI tools might be used to suggest protocols for pathogen testing, as described previously for physical processes. More consequentially, GPAI, BTs, and datasets could enable in silico testing of biological material.[14] Compared with in vivo or in vitro methods, in silico testing allows actors to gain insights on biological performance and characteristics quickly, safely, and without physical signatures that could be detected by intelligence or law enforcement entities.

Stage 6: Learn

We define *learn* as the analysis of in silico, in vivo, or in vitro testing to gain information to refine the pathogen. AI-enabled BTs, especially open-source tools and models trained on relevant sensitive biological data, could use experimental data from the testing step to suggest optimizations to refine the structure and function of a synthetic pathogen. This step is a key point at which the computational capabilities of AI models improve conventional experimentation to iterate through this part of the life cycle more quickly. As noted, this is an iterative process, and if the test and learn steps indicate that the pathogen is not functioning as desired, the actor can revert to the design step.

[11] Jonas B. Sandbrink, "Artificial Intelligence and Biological Misuse: Differentiating Risks of Language Models and Biological Design Tools," arXiv, arXiv:2306.13952v8, December 23, 2023.

[12] Jaspreet Pannu, Sarah L. Gebauer, Henry Alexander Bradley, Dulani Woods, Doni Bloomfield, Allison Berke, Greg McKelvey, Jr., Anita Cicero, and Tom Inglesby, *Defining Hazardous Capabilities of Biological AI Models: Expert Convening to Inform Future Risk Assessment*, RAND Corporation, CF-A3649-1, 2025.

[13] Bruce Wittmann, Tessa Alexanian, Craig Bartling, Jacob Beal, Adam Clore, James Diggans, Kevn Flyangolts, Bryan T. Gemler, Tom Mitchell, Steven T. Murphy, et al., "Toward AI-Resilient Screening of Nucleic Acid Synthesis Orders: Process, Results, and Recommendations," bioRxiv, December 4, 2024.

[14] National Academies of Sciences, Engineering, and Medicine, *Disseminating In Silico and Computational Biological Research: Navigating Benefits and Risks: Proceedings of a Workshop*, National Academies Press, 2025c.

Stage 7: Weaponization

We define *weaponization* as the stabilization and scaling of the material for dissemination. Historically, weaponization has been a difficult step for state and nonstate biological weapons programs, and the tacit knowledge required for succeeding at this step is generally very specific to bioweapons experience versus general biology expertise.[15] Aerosolization and hardening against sunlight, moisture, and other environmental conditions are difficult to optimize, and there is significant variability for different types of pathogens, release methods, and exposure routes. Multimodal GPAI inputs might provide instructions at this step, as well, and AI-enabled BTs—especially when left without guardrails—might introduce alterations that are key for weaponization. However, the step of weaponization may be less relevant when the pathogen is highly transmissible and does not require scale up or wide-spread dissemination for its effects to be realized.

[15] Ouagrham-Gormley, 2014.

APPENDIX C

Extended Approach

Our goal was to create a set of mitigations that would meaningfully reduce the risk of an AI-enabled biological incident and that could be readily implemented within the next two to five years with concerted efforts by relevant stakeholders in governments, the AI industry, academia, and the private sector. We sought to create a network of mitigations that were focused and concise, reasoning that effective implementation would be enabled by dedicated attention to a minimal viable set of targeted actions rather than numerous lines of effort. Although we aimed for a concise list of mitigations, we still prioritized making the mitigations robust and complementary such that, should one mitigation fail, others would still be capable of preventing the weaponization of AI-enabled biology in accordance with our outlined scope.

We first set out to identify and compile a list of potential mitigations. To do this, we reviewed academic and gray literature with a focus on a previous analysis. We started with a list of 67 potential mitigations, primarily consisting of those compiled by the authors of "Prioritizing Feasible and Impactful Actions to Enable Secure AI Development and Use in Biology" and integrating other examples from relevant literature.[1] These mitigations span various layers of defense, including ones aimed at AI models themselves as well as such broader interventions as technical model safeguards, materials access, user access, and data-access controls. Some mitigations were also focused on policy.

We then engaged in an iterative mitigation-refinement process. The process included several activities. The first was randomizing mitigations into distinct groups of five or ten to assess how the mitigations interacted with one another and what gaps became apparent with a limited number of interventions in play. Each member of the research team independently created ten iterations. We did this twice, once with the full list of mitigations broken into groups of five and once broken into groups of ten. This exercise helped us compare the relative strengths of the mitigations, note complementary mitigations, identify redundant mitigations, and determine which of the mitigations could be combined or streamlined.

Next, we assessed groups of mitigations across different types of actors, threats, targets, and methods of material and information acquisition. These included the following:

- **prototype actor type**—a low-resourced malicious individual, a radicalized expert scientist, a nonstate group, a terrorist group, and a state bioweapons program
- **material acquisition**—nature, lab stocks, core provider, nucleic acid provider who screens, nucleic acid provider who does not screen, black market, benchtop synthesizer, cloud labs, and contract research organizations
- **information acquisition**—closed-weight GPAI, open-weight GPAI, local GPAI, jailbreaking, biodesign tool, and the internet
- **threat type**—natural, enhanced, and novel
- **target type**—an airport or plane, a subway or metro, an outdoor mass gathering, an indoor mass gathering, and small-scale indoor exposure.

[1] Dettman et al., 2026.

These categories were meant to inform our thinking and to capture a broad variety of possible biological threats so we could understand how each mitigation would interact with these variables. Because of this, these categories might include variables that are outside the scope of our final mitigation strategy. To complete this exercise, we randomized both the list of mitigations under consideration and the variables of the biological threat scenario. We independently analyzed 30 iterations each, grouping seven mitigations at a time and comparing them against each randomized biological threat scenario. We then debated the strengths and weaknesses of each mitigation to select which network would advance to the next round of randomization.

We also assessed where the mitigations mapped across the AI-enabled biothreat risk pathway. After conducting these activities, we selected a preliminary network of ten mitigations to examine further. We then engaged in iterative team discussions and informal discussions with RAND experts to assess the strength of the ten mitigations and whether the list could be refined further. Ultimately, this led to the final list of nine mitigations present in this report.

Finally, we examined how different potential trajectories for policy, governance, and technical progress might affect the effectiveness and ease of implementing each mitigation. To do so, we assessed each mitigation against 13 future conditions, evaluating the mitigations' performance and robustness under each condition's potential outcome. The conditions were the following:

- model distribution
- biocapable model release
- compute availability or governance
- lab automation
- U.S.-China cooperation
- regulatory environment
- enforcement mechanisms
- catalyzing events
- international coordination
- public-private dynamics
- AI agent ecosystems
- AI progress
- AI bubble.

This analysis informed how we developed each mitigation's scope, technical specifics, and implementation requirements. It also shaped our assessment of each mitigation's technical feasibility and helped us understand how the mitigations function collectively to reduce risk across a variety of futures. We found that no single mitigation is sufficient—their value lies in how they layer across uncertain futures.

Abbreviations

AI	artificial intelligence
AI x Bio	the intersection of AI and biology
BT	biological tool
COVID-19	coronavirus disease 2019
DNA	deoxyribonucleic acid
GPAI	general-purpose AI
LLM	large language model
MCM	medical countermeasure
PPE	personal protective equipment
RNA	ribonucleic acid

References

AI Security Institute, *International AI Safety Report 2026*, February 2026.

Anthropic, "Introducing Anthropic's AI for Science Program," press release, May 5, 2025a.

Anthropic, "Introducing Claude Sonnet 4.5," press release, September 29, 2025b.

Attal-Juncqua, Aurelia, John P. Tarangelo, Evan Davison Kotler, Casey Aveggio, Marie Burns, Barbara Del Castello, Tyler Burns, Steph Guerra, and Saskia Popescu, *Building Resilience Against Artificial Intelligence–Enabled Biological Threats: Conference Proceedings from a RAND and Helena Workshop*, RAND Corporation, CF-A4954-1, 2026. As of May 28, 2026:
https://www.rand.org/pubs/conf_proceedings/CFA4954-1.html

Aveggio, Casey, Aman J. Patel, Sella Nevo, and Kyle Webster, *Exploring the Offense-Defense Balance of Biology: Identifying and Describing High-Level Asymmetries*, RAND Corporation, PE-A4102-1, August 2025. As of April 27, 2026:
https://www.rand.org/pubs/perspectives/PEA4102-1.html

Beaver, William, Yong-Bee Lim, Lillian Parr, Christine Parthemore, Andrew Weber, Natasha Bajema, Rohit Chitale, Jackson duPont, Chris Fall, Nikki Teran, and Alexander Titus, *A Handbook for Ending Catastrophic Biological Risks: How the United States Can Prevent Future Pandemics and Deter Biological Weapons*, ed. by Francesco Femia and Christine Parthemore, Janne E. Nolan Center on Strategic Weapons, Council on Strategic Risks, December 2021.

Berke, Allison, Forrest W. Crawford, Toby Webster, James Smith, Sana Zakaria, and Sella Nevo, *Data and AI-Enabled Biological Design: Risks Related to Biological Training Data and Opportunities for Governance*, RAND Corporation, PE-A3886-1, June 2025. As of March 23, 2026:
https://www.rand.org/pubs/perspectives/PEA3886-1.html

Biomni, homepage, undated. As of May 1, 2026:
https://biomni.stanford.edu/

Bipartisan Commission on Biodefense, *The National Blueprint for Biodefense: Immediate Action Needed to Defend Against Biological Threats*, April 2024.

Black, James R. M., Moritz S. Hanke, Aaron Maiwald, Tina Hernandez-Boussard, Oliver M. Crook, and Jaspreet Pannu, "Open-Weight Genome Language Model Safeguards: Assessing Robustness via Adversarial Fine-Tuning," arXiv, arXiv:2511.19299, November 24, 2025.

Bloomfield, Doni, James R. M. Black, Oliver Crook, Nadav Brandes, Moritz S. Hanke, Thomas V. Inglesby, Anita Cicero, Robert Pollack, Tina Hernandez-Boussard, Michael J. Imperiale, et al., "Biological Data Governance in an Age of AI," *Science*, Vol. 391, No. 6785, February 5, 2026.

Bo, Chen, Calvin Zhang, Christina Q. Knight, Nicholas Kruus, Jason Hausenloy, Pedro Medeiros, Nathaniel Li, Aiden Kim, Yury Orlovskiy, Coleman Breen, et al., "LLM Novice Uplift on Dual-Use, In Silico Biology Tasks," arXiv, arXiv:2602.23329, February 26, 2026.

Brady, Kyle, Jeffrey Lee, Dawid Maciorowski, Alyssa Worland, Jordan Despanie, Bria Persaud, Barbara Del Castello, Henry Alexander Bradley, Grant Ellison, Charles Teague, Sarah L. Gebauer, Greg McKelvey, Jr., Steph Guerra, and Ella Guest, *Bridging the Digital to Physical Divide: Evaluating LLM Agents on Benchtop DNA Acquisition*, RAND Corporation, RR-A4591-1, 2026. As of March 23, 2026:
https://www.rand.org/pubs/research_reports/RRA4591-1.html

Brent, Roger, and Greg McKelvey, Jr., *Contemporary Foundation AI Models Increase Biological Weapons Risk*, RAND Corporation, PE-A3853-1, December 2025. As of March 20, 2026:
https://www.rand.org/pubs/perspectives/PEA3853-1.html

Carter, Sarah R., *Developing a Customer Screening Framework for the Life Sciences*, Blueprint Biosecurity, March 2024.

Carter, Sarah R., and Greg Butchello, *A Framework for Managed Access to Biological AI Tools*, Nuclear Threat Initiative, January 28, 2026.

Casper, Stephen, Kyle O'Brien, Shayne Longpre, Elizabeth Seger, Kevin Klyman, Rishi Bommasani, Aniruddha Nrusimha, Ilia Shumailov, Sören Mindermann, Steven Basart, et al., "Open Technical Problems in Open-Weight AI Model Risk Management," *Transactions on Machine Learning Research*, March 2026.

Center for AI Standards and Innovation, *Evaluation of DeepSeek AI Models*, National Institute of Standards and Technology, 2025.

CEPI, "Preparedness in Practice: Rwanda's Record-Breaking Marburg Response," undated.

CEPI, "Securing the Pandemic Preparedness Engine," September 18, 2025.

Chen, Yadan, Mycal Tucker, Nina Panickssery, Tony Wang, Francesco Mosconi, Anjali Gopal, Carson Denison, Linda Petrini, Jan Leike, Ethan Perez, and Mrinank Sharma, "Enhancing Model Safety Through Pretraining Data Filtering," *Alignment Science Blog*, August 19, 2025. As of April 27, 2026: https://alignment.anthropic.com/2025/pretraining-data-filtering/

Cottier, Ben, Josh You, Natalia Martemianova, and David Owen, "How Far Behind Are Open Models?" EpochAI, November 4, 2024.

Danzig, Richard, *Preparing for Catastrophic Bioterrorism: Toward a Long-Term Strategy for Limiting the Risk*, Center for Technology and National Security Policy, May 2008.

Davies, Xander, Giorgi Giglemiani, Edmund Lau, Eric Winsor, Geoffrey Irving, and Yarin Gal, "Boundary Point Jailbreaking of Black-Box LLMs," arXiv, arXiv:2602.15001v2, February 18, 2026.

Del Castello, Barbara, and Henry H. Willis, *Assessing the Impacts of Technology Maturity and Diffusion on Malicious Biological Agent Development Capabilities: Demonstrating a Transparent, Repeatable Assessment Method*, RAND Corporation, RR-A3662-1, 2025. As of May 27, 2026: https://www.rand.org/pubs/research_reports/RRA3662-1.html

Dettman, Josh, Emily Lathrop, Aurelia Attal-Juncqua, Matthew Nicotra, and Allison Berke, "Prioritizing Feasible and Impactful Actions to Enable Secure AI Development and Use in Biology," *Biotechnology and Bioengineering*, Vol. 123, No. 5, May 2026.

Dev, Sunishchal, Charles Teague, Grant Ellison, Kyle Brady, Ying-Chiang Jeffrey Lee, Sarah L. Gebauer, Henry Alexander Bradley, Dawid Maciorowski, Bria Persaud, Jordan Despanie, Barbara Del Castello, Alyssa Worland, Michael Miller, Adrian Salas, Dave Nguyen, James Liu, Jason Johnson, Andrew Sloan, Will Stonehouse, Travis Merrill, Thomas Goode, Greg McKelvey, Jr., and Ella Guest, *Toward Comprehensive Benchmarking of the Biological Knowledge of Frontier Large Language Models*, RAND Corporation, RR-A3797-1, 2025. As of April 27, 2026: https://www.rand.org/pubs/research_reports/RRA3797-1.html

Drexel, Bill, and Caleb Withers, *AI and the Evolution of Biological National Security Risks*, Center for a New American Security, August 2024.

Executive Order 14292, "Improving the Safety and Security of Biological Research," Executive Office of the President, May 5, 2025.

Feldman, Jonathan, Tal Feldman, and Annie I. Anton, "Know Your Scientist: KYC as Biosecurity Infrastructure," *Frontiers in Microbiology*, Vol. 17, April 29, 2026.

Gerstein, Daniel M., *Bioterror in the 21st Century: Emerging Threats in a New Global Environment*, Naval Institute Press, 2009.

Götting, Jasper, Pedro Medeiros, Jon G. Sanders, Nathaniel Li, Long Phan, Karam Elabd, Lennart Justen, Dan Hendrycks, and Seth Donoughe, "Virology Capabilities Test (VCT): A Multimodal Virology Q&A Benchmark," arXiv, arXiv:2504.16137v2, April 29, 2025.

Groff-Vindman, Cyndy S., Benjamin D. Trump, Christopher L. Cummings, Madison Smith, Alexander J. Titus, Ken Oye, Valentina Prado, Eyup Turmus, and Igor Linkov, "The Convergence of AI and Synthetic Biology: The Looming Deluge," *npj Biomedical Innovations*, Vol. 2, July 1, 2025.

Jeffery, Nazish, Sarah R. Carter, Tessa Alexanian, Oliver Crook, Samuel Curtis, Richard Moulange, Shrestha Rath, Sophie Rose, and Jennifer Clarke, "Bio x AI: Policy Recommendations for a New Frontier," Federation of American Scientists, December 12, 2023.

Kelland, Kate, "The 100 Days Mission—How Close Are We?" CEPI, July 24, 2024.

Lee, Ying-Chiang Jeffrey, Bria Persaud, Barbara Del Castello, Allison Berke, and Gustavs Zilgalvis, *Documenting Cloud Labs and Examining How Remotely Operated Automated Laboratories Could Enable Bad Actors*, RAND Corporation, PE-A3851-1, April 2025. As of May 28, 2026: https://www.rand.org/pubs/perspectives/PEA3851-1.html

Lingtao Chen, Qiaomu Li, Kazi Fahim Ahmad Nasif, Ying Xie, Bobin Deng, Shuteng Niu, Seyedamin Pouriyeh, Zhiyu Dai, Jiawei Chen, and Chloe Y. Xie, "AI-Driven Deep Learning Techniques in Protein Structure Prediction," *International Journal of Molecular Sciences*, Vol. 25, No. 15, August 1, 2024.

Luckey, David, Sara Duhachek Muggy, Taylor Frey, David Stebbins, Tracey Rissman, Bianca Espinosa, Daniel Tapia, Greg McKelvey, Jr., Neeti Pokhriyal, Joseph Dawson, Sara Hughes, Morgan Sandler, Rushil Bakhshi, Marta Kepe, Geoffrey Kirkwood, Sarah W. Denton, David DeSmet, Minami Makino, Ella Guest, Sina Beaghley, Suzanne Genc, Michael Miller, Skye A. Miner, Barbara Del Castello, Forrest W. Crawford, Jeffrey Lee, Clay Strickland, Sunny D. Bhatt, John Vahedi, Lydia Grek, Vanya Barrer, Ramiro Insuasti, Jr., Jack Lashendock, Derek Roberts, Aleksandr Esparza Hartunian, Shannon Walsh, Will Shumate, Elliott Brennan, Tyler Liggett, Kara Jia, Ajay K. Kochhar, James Smith, and James Ryseff, *Mitigating Risks at the Intersection of Artificial Intelligence and Chemical and Biological Weapons*, Homeland Security Operational Analysis Center operated by the RAND Corporation, RR-A2990-1, 2025. As of March 20, 2026: https://www.rand.org/pubs/research_reports/RRA2990-1.html

Maslej, Nestor, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Njenga Kariuki, Emily Capstick, Anka Reuel, Erik Brynjolfsson, John Etchemendy, et al., *The AI Index 2025 Annual Report*, AI Index Steering Committee, Institute for Human-Centered AI, Stanford University, April 2025.

Mitchener, Ludovico, Angela Yiu, Benjamin Chang, Mathieu Bourdenx, Tyler Nadolski, Arvis Sulovari, Eric C. Landsness, Daniel L. Barabasi, Siddharth Narayanan, Nicky Evans, et al., "Kosmos: An AI Scientist for Autonomous Discovery," arXiv, arXiv:2511.02824, November 4, 2025.

National Academies of Sciences, Engineering, and Medicine, "AI-Enabled Biological Design and the Risks of Synthetic Biology," in *The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations*, National Academies Press, 2025a.

National Academies of Sciences, Engineering, and Medicine, "Design-Build-Test-Learn: Impact on the Synthetic Biology Process," in *The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations*, National Academies Press, 2025b.

National Academies of Sciences, Engineering, and Medicine, *Disseminating In Silico and Computational Biological Research: Navigating Benefits and Risks: Proceedings of a Workshop*, National Academies Press, 2025c.

National Institute of Standards and Technology, "CAISI Evaluation of DeepSeek AI Models Finds Shortcomings and Risks," press release, last updated November 20, 2025.

Navigating the Benefits and Risks of Publishing Studies of In Silico Modeling and Computational Approaches of Biological Agents and Organisms—A Workshop, National Academies of Sciences, Engineering, and Medicine, April 3–4, 2025.

O'Brien, Kyle, Stephen Casper, Quentin Anthony, Tomek Korbak, Robert Kirk, Xander Davies, Ishan Mishra, Geoffrey Irving, Yarin Gal, and Stella Biderman, "Deep Ignorance: Filtering Pretraining Data Builds Tamper-Resistant Safeguards into Open-Weight LLMs," arXiv, arXiv:2508.06601, August 8, 2025.

Office of the Director of National Intelligence, "JCAT Counter Terrorism Guide for Public Safety Personnel," webpage, undated. As of March 20, 2026: https://www.dni.gov/nctc/jcat/index.html

Office of the President of the United States, *National Policy on the Transfer of Scientific, Technical and Engineering Information*, NSDD-189, September 21, 1985.

Office of the President of the United States, *National Biodefense Strategy*, 2018.

Office of the President of the United States, *National Biodefense Strategy and Implementation Plan for Countering Biological Threats, Enhancing Pandemic Preparedness, and Achieving Global Health Security*, October 2022.

OpenAI, "Preparing for Future AI Capabilities in Biology," June 18, 2025.

OpenAI, "Strengthening Societal Resilience with Rosalind Biodefense," May 29, 2026.

Ouagrham-Gormley, Sonia Ben, *Barriers to Bioweapons: The Challenges of Expertise and Organization for Weapons Development*, Cornell University Press, 2014.

Pannu, Jaspreet, Doni Bloomfield, Robert MacKnight, Mortiz S. Hanke, Alex Zhu, Gabe Gomes, Anita Cicero, and Thomas V. Inglesby, "Dual-Use Capabilities of Concern of Biological AI Models," *PLoS Computational Biology*, Vol. 21, No. 5, May 8, 2025.

Pannu, Jaspreet, Sarah L. Gebauer, Henry Alexander Bradley, Dulani Woods, Doni Bloomfield, Allison Berke, Greg McKelvey, Jr., Anita Cicero, and Tom Inglesby, *Defining Hazardous Capabilities of Biological AI Models: Expert Convening to Inform Future Risk Assessment*, RAND Corporation, CF-A3649-1, 2025. As of March 23, 2026:
https://www.rand.org/pubs/conf_proceedings/CFA3649-1.html

Righetti, Luca, Kamile Lukosiute, and James Black, "Coding Agents Are Changing the Biosecurity Risk Landscape," *GovAI* blog, April 20, 2026. As of May 1, 2026:
https://www.governance.ai/analysis/coding-agents-are-changing-the-biosecurity-risk-landscape

Rissman, Tracey, and Annette Prieto, *Attributing Biological Weapons Use: Strengthening Department of Defense Capabilities to Investigate Deliberate Biological Incidents*, RAND Corporation, RR-A2360-1, 2024. As of March 20, 2026:
https://www.rand.org/pubs/research_reports/RRA2360-1.html

Rose, Sophie, and Cassidy Nelson, *Understanding AI-Facilitated Biological Weapon Development*, Centre for Long-Term Resilience, October 2023.

Sandbrink, Jonas B., "Artificial Intelligence and Biological Misuse: Differentiating Risks of Language Models and Biological Design Tools," arXiv, arXiv:2306.13952v8, December 23, 2023.

Shilov, Igor, Alex Cloud, Aryo Pradipta Gema, Jacob Goldman-Wetzler, Nina Panickssery, Henry Sleight, Erik Jones, and Cem Anil, "Beyond Data Filtering: Knowledge Localization for Capability Removal in LLMs," arXiv, arXiv:2512.05648, December 5, 2025.

U.S. Army, "Army Antiterrorism Officer Ready Reference: Vigilance," undated.

U.S. Department of Health and Human Services, Administration for Strategic Preparedness and Response, *Screening Framework Guidance for Providers and Users of Synthetic Nucleic Acids*, October 2023.

Washington, Michael A., Joubert Paulino, and Siang-Chean Kua, "Barriers to Biological Weapons Development: Potential Implications for Pathway Disruption," *Countering WMD Journal*, No. 28, Spring/Summer 2024.

Webster, Toby, Richard Moulange, Barbara Del Castello, James Walker, Sana Zakaria, and Cassidy Nelson, *Global Risk Index for AI-Enabled Biological Tools*, Centre for Long-Term Resilience, September 9, 2025.

Wei, Boyi, Zora Che, Nathaniel Li, Udari Madhushani Sehwag, Jasper Götting, Samira Nedungadi, Julian Michael, Summer Yue, Dan Hendrycks, Peter Henderson, et al., "Best Practices for Biorisk Evaluations on Open-Weight Bio-Foundation Models," arXiv, arXiv:2510.27629, October 31, 2025.

Williams, Adeline E., Barbara Del Castello, Jeffrey Lee, Derek Roberts, John P. Tarangelo, Jay Atanda, Alejandro Colman-Lerner, Jeff Gerold, and Roger Brent, *Developing a Risk-Scoring Tool for Artificial Intelligence–Enabled Biological Design: A Method to Assess the Risks of Using Artificial Intelligence to Modify Select Viral Capabilities*, RAND Corporation, RR-A4490-1, 2026. As of May 27, 2026:
https://www.rand.org/pubs/research_reports/RRA4490-1.html

Willis, Henry H., Adeline E. Williams, Saskia Popescu, Derek Roberts, Eva Coringrato, Laura J. Faherty, Pedro Nascimento de Lima, and Sana Zakaria, "A Framework for Evaluating Implementation, Impact, and Cost-Effectiveness of Wastewater and Environmental Surveillance," *Frontiers in Public Health*, Vol. 14, March 18, 2026.

Wittmann, Bruce, Tessa Alexanian, Craig Bartling, Jacob Beal, Adam Clore, James Diggans, Kevn Flyangolts, Bryan T. Gemler, Tom Mitchell, Steven T. Murphy, et al., "Toward AI-Resilient Screening of Nucleic Acid Synthesis Orders: Process, Results, and Recommendations," bioRxiv, December 4, 2024.

Wittmann, Bruce, Tessa Alexanian, Craig Bartling, Jacob Beal, Adam Clore, James Diggans, Kevin Flyangolts, Bryan T. Gemler, Tom Mitchell, Steven T. Murphy, et al., "Strengthening Nucleic Acid Biosecurity Screening Against Generative Protein Design Tools," *Science*, Vol. 390, No. 6768, October 2, 2025.

Zhang, Kang, Xin Yang, Yifei Wang, Yunfang Yu, Niu Huang, Gen Li, Xiaokun Li, Joseph C. Wu, and Shengyong Yang, "Artificial Intelligence in Drug Development," *Nature Medicine*, Vol. 31, January 2025.

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